

Notes on Applications and Demo ... beyond the ML course toward research

DRAFT, please do not circulate!

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2025



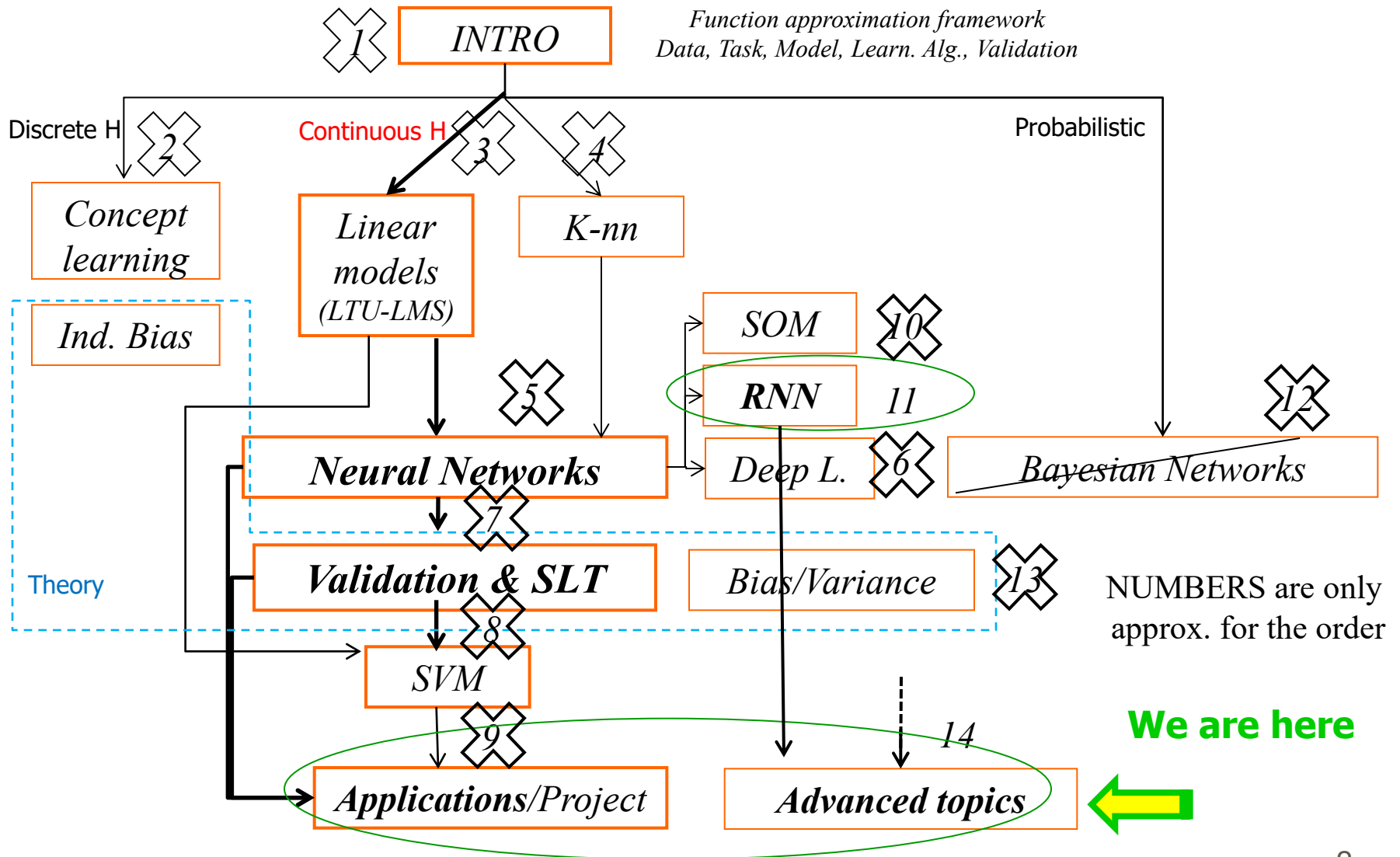
Dipartimento di Informatica
Università di Pisa - Italy

ML Course structure

Where we are and go



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Program (page 1)

- **Introduction:**
 - Critical contextualization in CS, learning as function approximation (example)
 - Computational learning tasks for predictions, design components, hypothesis space, model, learning algorithm, Loss and learning tasks.
- **Basic concepts and models:**
 - Concept Learning, dimension of discrete spaces, version space, Inductive Bias (language and search bias) and its role.
 - Complexity over an example (polynomials), Generalization and overfitting concepts, inductive principles, introduction to SLT and validation
 - Linear models: LTU, LMS algorithms (direct and iterative solutions): development and limitations. Linear basis expansion (LBE). Regularization
 - K-Nearest Neighbors: algorithm, characteristics, comparison, curse of dimensionality

Topics are NOT ordered according to the lecture number

Program (page 2)

- **Neural Network (NN) models:** architectures and learning algorithms
 - Biological and Artificial Neural Networks, Perceptron and LTU, Perceptron learning algorithms, Adaline, computational properties and comparison
 - Perceptron convergence theorem. Activation functions, MLP and NN architectures, NN properties: NN as LBE, universal approximation.
 - Derivation of the Back-propagation algorithm for learning in MLP
 - Hyper parameters and heuristics for the NN learning. Issues and techniques to improve the training (e.g. W init. and multiple minima, momentum, minibatch, learning rate setting and variants, ...). Early stopping and Regularization. NN constructive approaches: Cascade Correlation.
 - Discussion on the characteristics of NNs.
 - Applicative examples (NN specific architecture design): Convolutional networks
 - Deep Learning: advantages and flattening analysis, autoencoders, representation learning and distributed learning concepts, other insights, issues and specific techniques including ReLu, Drop-out, pre-training, etc...
 - Randomized NN (with Cover Theorem)
 - NN architectures: Recurrent Neural Networks (intro)

Topics are NOT ordered according to the lecture number

Program (page 3)

- **Principles of learning processes and general practical aspects:**
 - Evaluation (Risk estimation) and Model Selection: difference and combined use, index (error measure) for classification and regression
 - Analytical or re-sampling approaches. Hold-out, K-fold Cross-validation schemas. Special cases, typical errors.
 - Statistical Learning Theory (SLT): complexity and shattering, VC-dimension (examples), basic SLT/VC-bound, Structural Risk Minimization
 - Bias-Variance decomposition. Outline of ensemble learning (bagging, boosting)
- **Support Vector Machines and Kernel-based models:**
 - Linear model and margin (hard margin), Vapnik theorem, soft-margin
 - Non-linear tasks: Kernels.
 - SVM for regression.
 - Critical analysis of characteristics and hyperparameters, discussion and comparisons.

Topics are NOT ordered according to the lecture number

Program (page 4)

- **Unsupervised learning:**
 - Usage, Vector Quantization, Self-Organizing Maps
- **Introduction to the Applications:** benchmark examples, applicative project
- **Introduction to advanced approaches:** Learning for Structured data (learning on graphs)
- **Introduction to the Applications:** examples (and research demos)

Topics are NOT ordered according to the lecture number

Program (page 5): Summary



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Beyond the model details and techniques,
as transversal concepts should be emerge:

- The relevance and roles of *model complexity* and *regularization for generalization* purposes
- The relevance and roles of a proper, extensive and rigorous *model selection (validation) and assessment*
- That **research in ML is open, and you are welcome to join !**



*“tie a knot in your
handkerchief”*

Future – just ahead



- General topics for ML applications
- Didactics: CNS course
- CIML@Pisa
- CIML@Pisa research examples

ML: continuously growing area
in the modeling-theoretical and applied aspects
(e.g. frontier of scientific research)

A more general look to applications

General challenges

- Build autonomous Intelligent/learning machines:
 - Robotics, HRI, search engines, ...
- Build powerful tools for emerging challenges in intelligent data analysis
 - Tools for the “data scientist”
- Open new areas of applications in CS: innovative interdisciplinary open problems
 - Fantasy is your limitation !

Challenges by Source of data

2000's

- **Bio-Cheminformatics** (*-omics, personalized medicine, toxicology)
- **Web** data

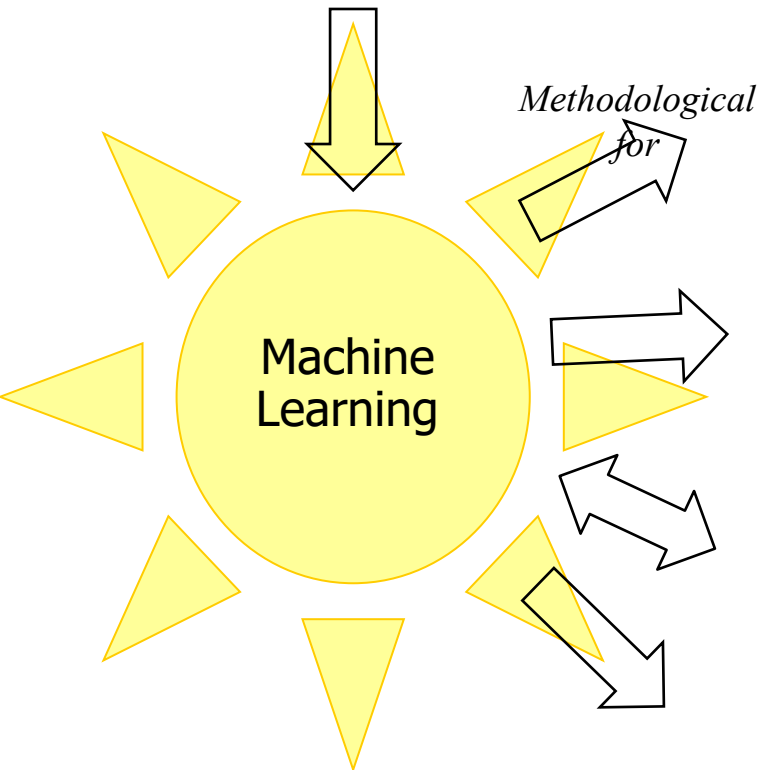
Recent "buzzwords"

- **Smart** systems (design smart things and make human life better) : from *Smart Cities* to *Ambient Assisted Living* to the *Internet of Things* (One trillion = 10^{12} devices with sensors of all kinds are expected to be connected to the Internet by 2022 (IEEE news March 2014))
- **Social, Big data** : e.g. social network data, or again IoT

Growing data sources opens up a huge application area for ML and related areas (also in terms of employment opportunities)

ML & other areas

Computational mathematics for learning and data analysis
Parallel and distributed systems: paradigms and models



Human language technologies (AI)
Generative and deep learning (AI)
Computer vision (AI)

Computational neuroscience (AI/ING)
Robotics (AI)
Semantic web (AI)

Algorithm engineering (KD)
Data mining (KD)
Information Retrieval (KD)
Mobile and cyber-physical systems (ICT)

See (link):

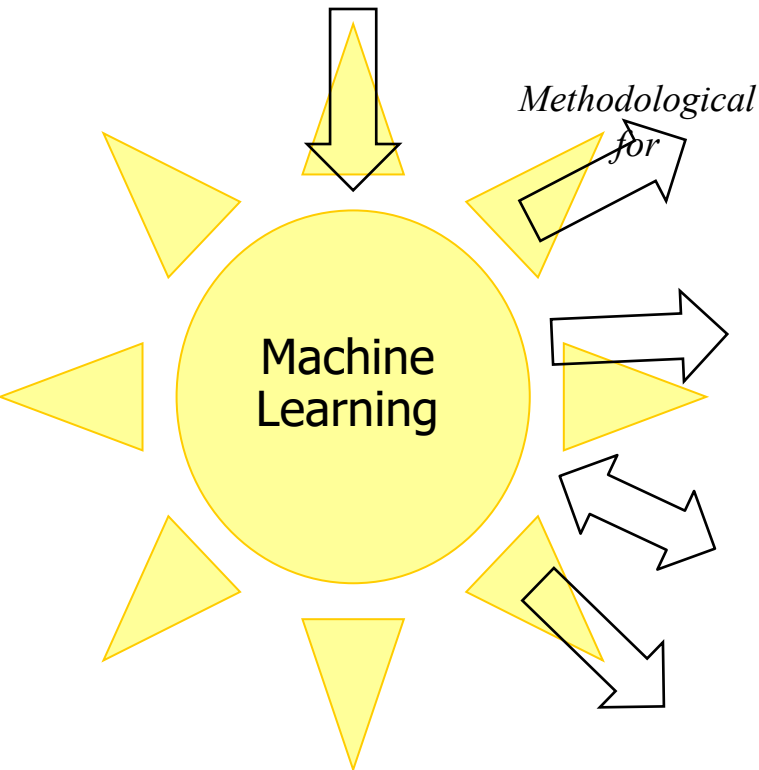
Artificial
Intelligence

But also others:*

Bioinformatics (KD) Advanced databases (KD)
Big data analytics (WBI) ...

ML & other areas

Computational mathematics for learning and data analysis
Parallel and distributed systems: paradigms and models



Human language technologies (AI)
Generative and deep learning (AI)
Computer vision (AI)

Computational neuroscience (AI/ING)

Robotics (AI)
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But also others:*

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Big data analytics (WBI) ...

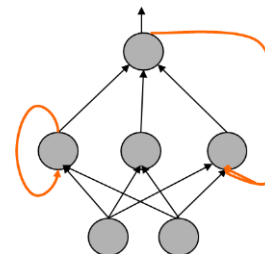
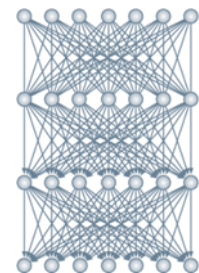
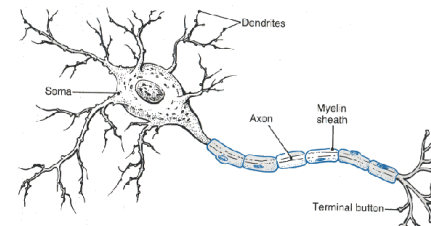
CNS - General Info

- **CNS (Computational neuroscience)** 6 CFU SSD: INF/01 is part of *Applied Brain Science (12 CFU)* - Master programme in Bionics Engineering
- **CNS** (6 CFU) - Master programme in Computer Science
- **Instructors (2024):**
 - ▶ ~~Alessio Micheli~~
 - ▶ Claudio Gallicchio
 - ▶ Andrea Ceni
 - ▶ Seminars from experts in CNS



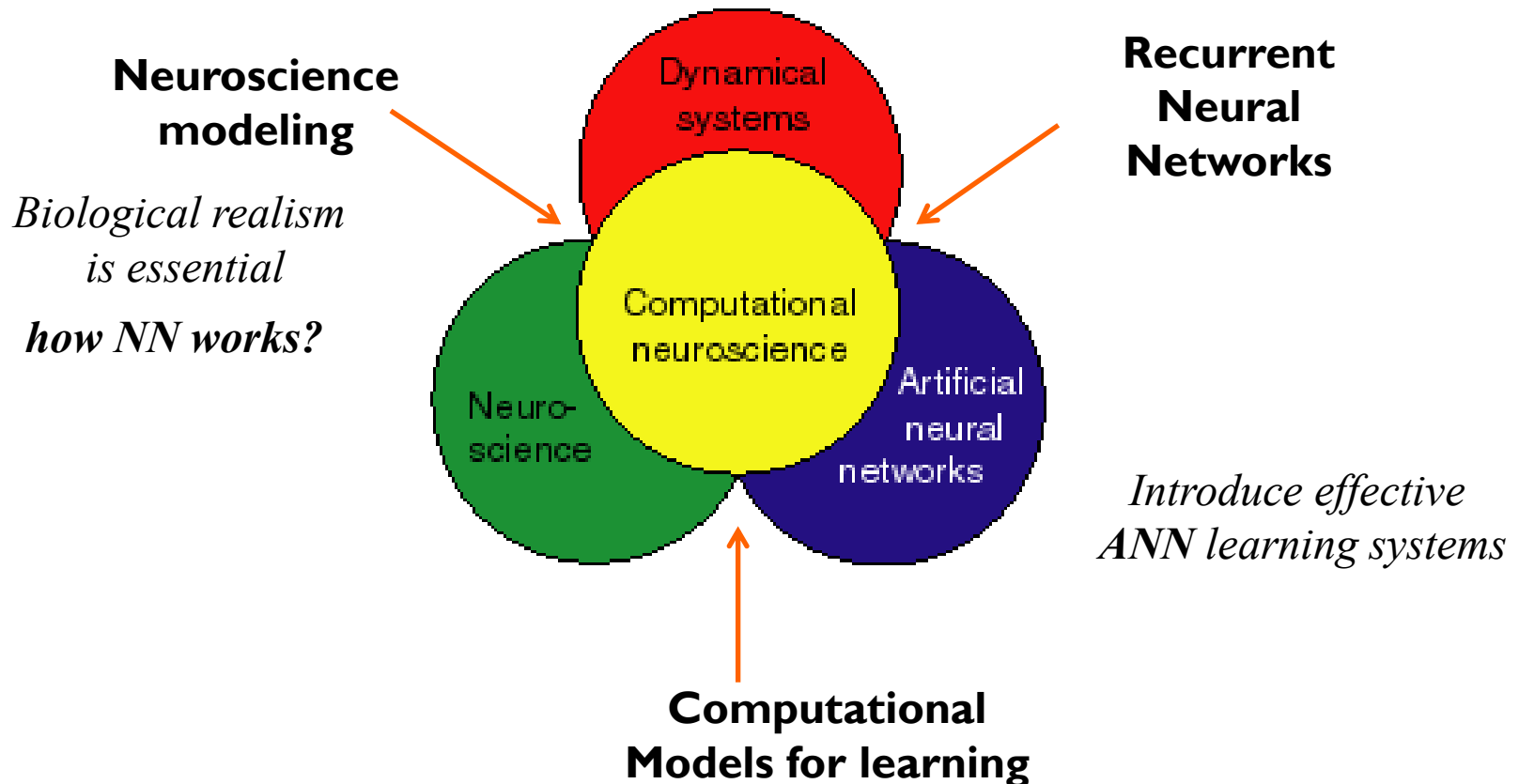
Objectives CNS

- Introduction to the basic knowledge of the CNS, considering both the **bio-inspired neural** modelling and **computational** point of view.
- Gain practical knowledge on simple CNS models by lab experience
- Including, as for Syllabus,
 - Bio-inspired neural modelling
 - Neuroscience modeling
 - E.g. spiking neuron models and synaptic plasticity
 - Advanced computational learning models:
 - Unsupervised learning: Hebbian and Hopfield Networks
 - Recurrent neural networks
 - Dynamical models for temporal data
 - Also Reservoir Computing



Our approach to CNS

Study of the information processing properties of the structures involved in the nervous system dynamics



CNS Q&A



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- **Attendance:** strongly advised. We develop the LAB in the class time, through Hands-on Labs in Python and MATLAB, *interleaving* theoretical and practical sessions.
- **Where?** “Polo Ingegneria” Via Giunta Pisano, in a lab class!
- **Are you ready?** **Yes**, for the ML and Math background, and then:
 - Python and MATLAB are easy to be learnt
 - (simple) differential equations for the Neuronal Dynamics Modelling (first part) will be gently introduced for the practical usage (CS point of view)
- The **Exam** include a *seminar* or a project + Oral (details will be provided)
- Further info: “unipi esami” portal for the programme

Future – just ahead



- General topics for ML applications
- Didactics: CNS course
- **CIML@Pisa**
- **CIML@Pisa research examples**

CIML Group - People

- **Faculty** (9): Prof. A. Micheli (coordinator), Prof. D. Bacciu, Prof. C. Gallicchio, Dr. A. Carta, Dr. A. Ceni, Dr. A. Cossu, Dr. L. Passaro, Dr. M. Podda, Dr. D. Tortorella
- 4+ post-doc researchers
- **12+ Phd students (the young energy!)**
- More than 25 years of research in ML/AI
- Part of the [CAIRNE.eu](https://cairne.eu) Research Network  (Confederation of Laboratories for Artificial Intelligence Research in Europe)



**Computational Intelligence &
Machine Learning Group**

<https://ciml.di.unipi.it/>

CIML Group - People

More than 10 PhD students ...



New calls each year, for

- PhD positions in **Computer Science** or
- the **National** PhD in **Artificial Intelligence**



*“tie a knot in
your
handkerchief”*



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Machine Learning Group**

<https://ciml.di.unipi.it/>

Motivations to a look at Ongoing CIML Projects

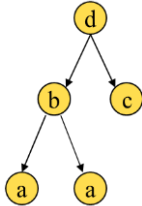
- To have a touch on *research on ML* @ di.unipi
- Some examples of projects @ **CIML-Pisa**
- On request + Motivational: ***University: didactics & research***



CIML Research Area (overview)



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- Learning in Structured Domains (SD)
 - Pioneering since the 90's the development of *Recursive NN* for **trees** and **graphs**
 - New models by RC, Kernels and generative approaches
 - Focus on efficient deep learning for SD
- Intelligent sensor networks (ML+WSN/IoT)   
 - FP7: Rubicon (self-learning robot ecologies), Doremi (personalized healthcare/monitoring of human daily activities)
 - H2020: TEACHING (adaptive CPSoS) [2020-2022], TAILOR (Foundations of Trustworthy AI - Integrating Reasoning, Learning and Optimization) [2020-2023], EMERGE [2023-2026] ...
 - NextGeneration EU programme: FAIR [2023-2025], THE [2022-2024], ...
 - BrAID (Brugada syndrome and Artificial Intelligence applications to Diagnosis)
- Interdisciplinary applications (med., bio., chem., eng., robotics, ...)
- Industrial collaborations
- ...



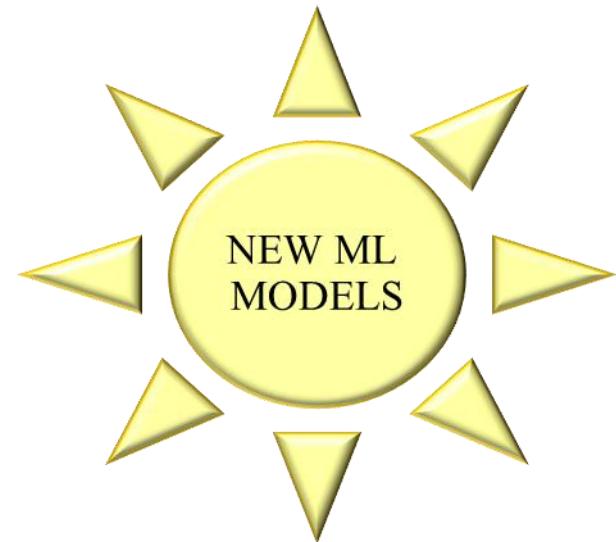
**Computational Intelligence &
Machine Learning Group**

<https://ciml.di.unipi.it/>

Main core: new model developments

Vertical approach to ML:
Theoretical analysis
New models development
Applications

- **Neural Networks**
- Reservoir Computing
- Kernel-based
- Generative-probabilistic
- **Deep learning**



See the previous lecture slides on SDL

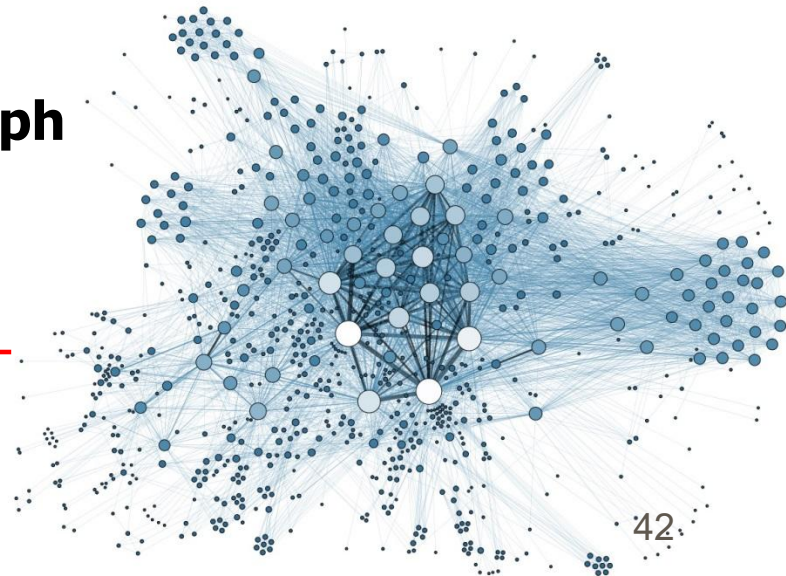
An example: an open area of research @CIML-Pisa



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- Deep Learning for SD (the *research core*!)
 - **Analysis of properties** of *Recurrent Neural Networks* with *deep* networks
 - **New models** to conjugate efficient RNN modeling and **deep** learning aspects/advantages.
 - **New kernels** from generative or dynamical models
 - Extension of ML to **Structured/Graph Domains** and **applications**

See the previous lecture slides on SDL



Interdisciplinary applications

Vertical approach to ML:

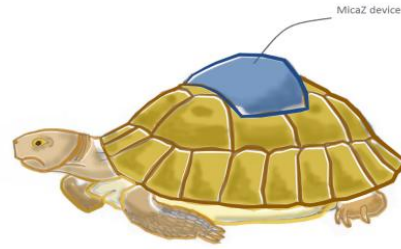
- **Theoretical analysis**
- **New models development**
 - (see previous lecture learning for SD)
- **Applications**

Application Overview

@CIML



Chem/Bio-informatics



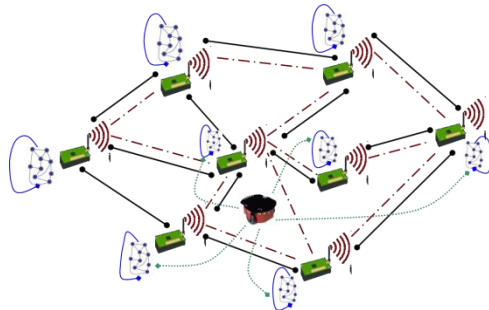
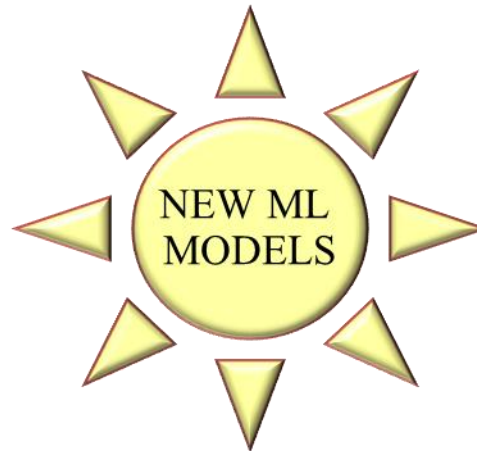
Pattern recognition on
animal activity



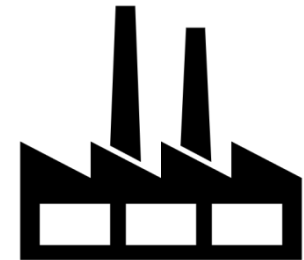
Sensor data
Smart *
IoT



Health care (projects)



Intelligent Sensor Networks (EU project)



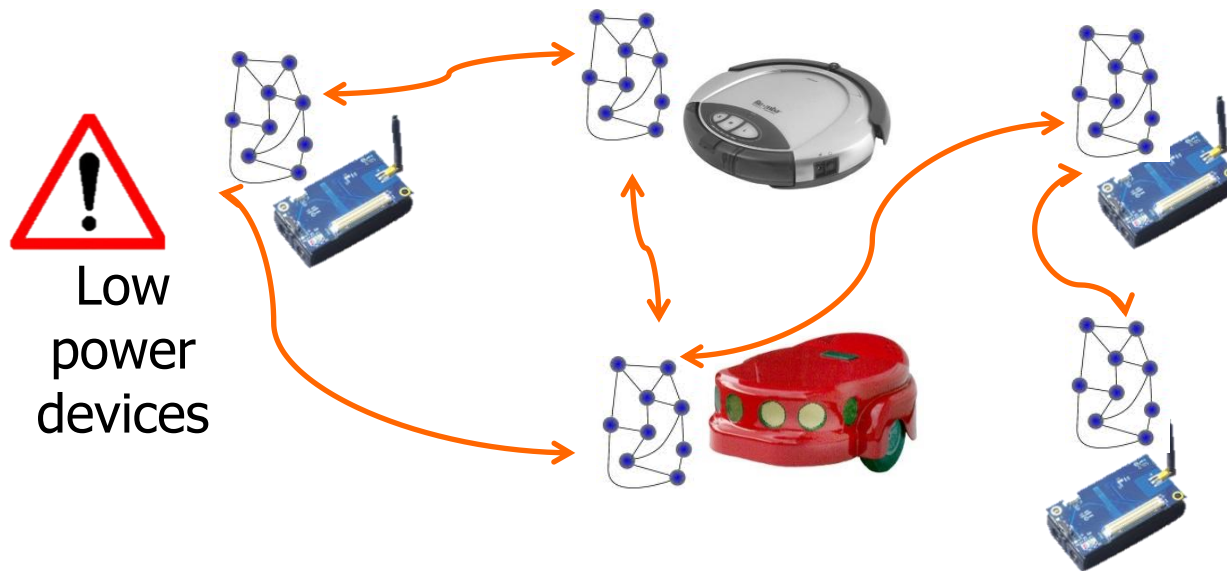
Data analysis for
industrial leaders

BACK 
TO THE PAST

PAST PROJECTS
Examples

A Learning System for Robotic Ecologies

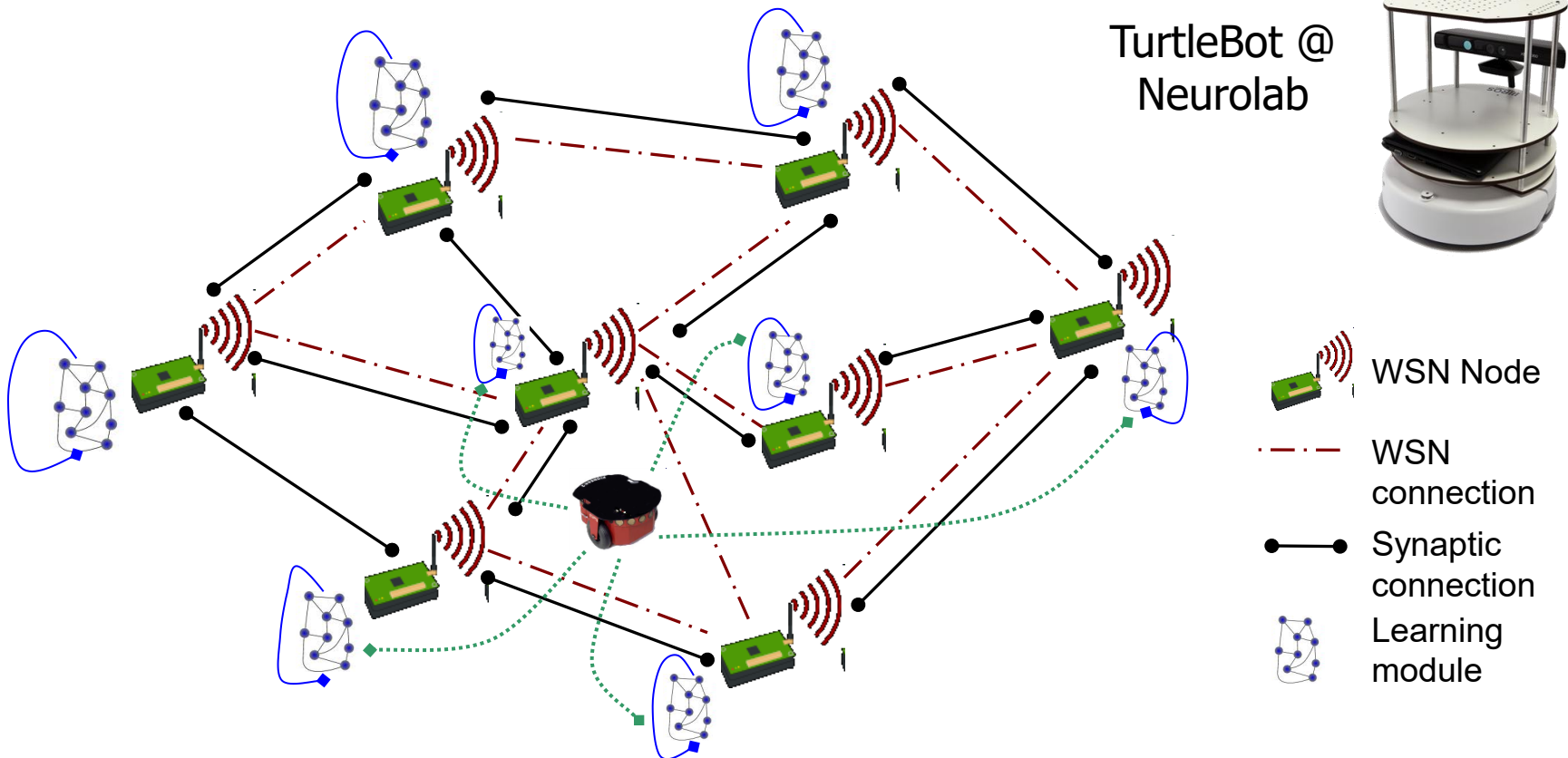
A distributed, adaptive, and self-organizing learning system comprising independent learning modules residing on ecology nodes



Why embedding?
Information fusion
Adaptive filtering
Energy saving

Learning modules interact to realize a distributed prediction

- Best with CNS/ISPR (and even better with Sensor Network course)
- Dynamical models for Wireless Sensor Networks data







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Reti robotiche per assistere gli anziani e i disabili

L'Università di Pisa partner del progetto di ricerca europeo Rubicon finanziato con 2,5 milioni di euro



Si chiama RUBICON ed è un robot capace di dare assistenza ad anziani e disabili dopo averne imparato le abitudini di vita. Un team di ricercatori del dipartimento di Informatica dell'Università di Pisa, coordinati da Stefano Chessa e Alessio Micheli, è partner di un progetto di ricerca europeo finanziato con 2,5 milioni di euro che ha l'obiettivo di unire dispositivi robotici e reti di sensori in un sistema capace di interagire con l'ambiente circostante. Della ricerca, il cui capofila in Europa è l'Università di Dublino, fanno parte anche l'Istituto di scienza e tecnologia dell'informazione "A. Faedo" (ISTI) del CNR, la Fondazione Stella Maris, la Örebro University (Svezia), la University of Ulster (Nord Irlanda), due aziende spagnole di componenti robotici (Tecnalia e Robotnik) e l'azienda irlandese Pintail Ltd.

L'idea che sta alla base del progetto RUBICON (Robotic Ubiquitous Cognitive Network) è progettare robot in grado di apprendere le caratteristiche dell'ambiente e della qualità di vita della persona da assistere tramite una rete di sensori installati in casa. RUBICON è infatti un misto di "intelligenza" ambientale e robotica, una sorta di "cervello artificiale pervasivo" che fa parte dell'ambiente ed è capace di apprendere dall'ambiente stesso ciò che avviene e di comandare i robot di conseguenza. I sensori acquisiscono le informazioni, memorizzano un modello di abitudini ritagliato sulla persona specifica a cui portare assistenza e lo trasmettono ai robot.

I ricercatori spiegano che, con questa tecnologia, sarà possibile abbinare un evento a un comportamento specifico del robot. Per fare un esempio, se RUBICON capisce che la persona ha l'abitudine di alzarsi tutte le mattine alle sette, il sistema attiva i robot a quell'ora più per portare le pantofole e assistere il soggetto specifico. Il vantaggio è che, se col tempo le abitudini cambiano, RUBICON è in grado di adattarsi autonomamente alle condizioni ambientali mutate, senza bisogno di essere riprogrammato.

Il team di ricerca dell'Università di Pisa, che include i gruppi di reti wireless e di intelligenza computazionale, è composto da 6 persone tra professori, ricercatori, assegnisti di ricerca e collaboratori, e ha lavorato negli ultimi sei mesi per definire il progetto che adesso passerà alla fase più operativa. RUBICON arriverà, infatti, a test di prototipi legati a scenari applicativi con una sperimentazione di sei mesi all'ospedale della Stella Maris e, parallelamente, in un living lab dell'azienda Tecnalia a Bilbao.



Nel primo caso è previsto l'utilizzo del robot per il trasporto di materiale ospedaliero: i sensori ambientali dovranno essere addestrati a memorizzare dati utili a comprendere e disegnare il percorso per arrivare ai degeniti. Il laboratorio spagnolo, invece, riprodurrà l'ambiente di un appartamento dove attori simuleranno azioni quotidiane sotto l'osservazione di sensori. Al suo lato sarà attrezzata una sala regia, dove i ricercatori potranno elaborare i dati, osservando i robot e la loro interazione con le persone.

Lo sviluppo di questa tecnologia apre nuove prospettive per il futuro: la sfida è riuscire a progettare sistemi che, in virtù della loro capacità di adattarsi alle persone, siano in grado di capire quello di cui hanno veramente bisogno, portando assistenza dove e quando serve, compatibilmente con la sensibilità del soggetto assistito, imparando a interagire con lui in maniera discreta e non invasiva.

Ne hanno parlato:

Ansa
Unità Toscana
Nazione Pisa
Corriere Fiorentino
Nuovo Corriere di Firenze
RepubblicaFirenze.it
CorriereFiorentino.it
TirrenoPisa.it
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A. Micheli

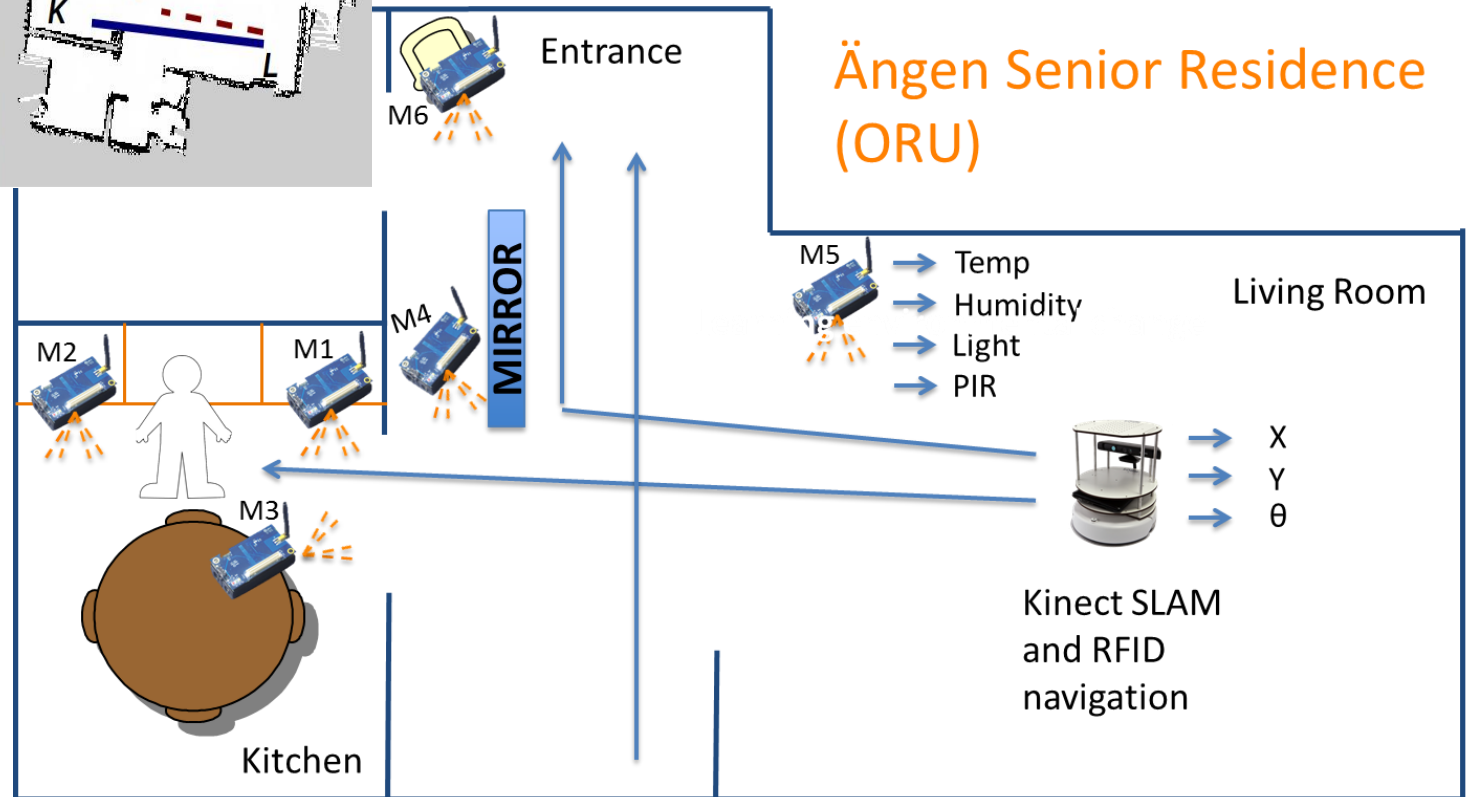
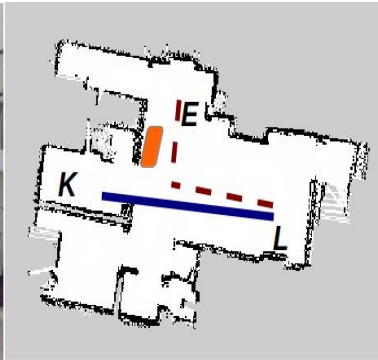
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Examples of applicative scenarios: Robot Navigation



Robot navigation in Stella Maris Hospital
Using noisy signals from RSSI Anchors

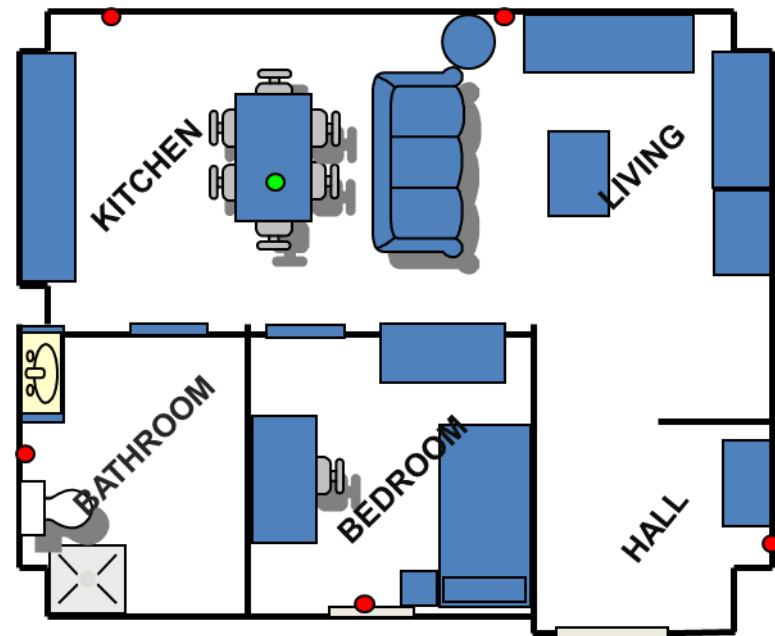
Examples of applicative scenarios: Learning for Robot Ecology



Learning/ adaptation to user's preferences and changes to the environment 51

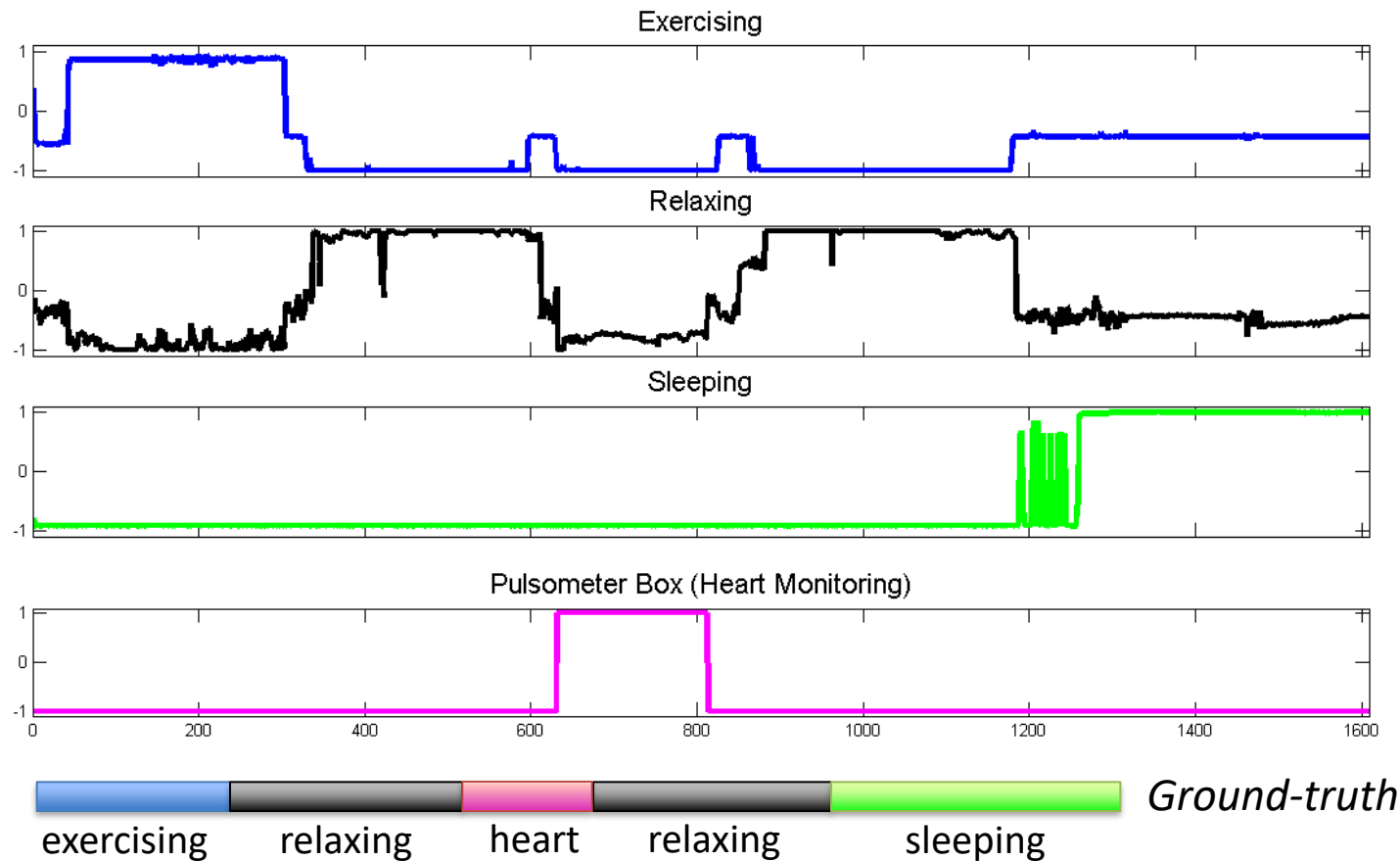
Examples of applicative scenarios: Ambient Assisted Living

- Predicting **event occurrence** and confidence of Human activities (from cooking to sleeping) basing on local sensors



AAL scenario at TECNALIA HomeLab
(Bilbao, Spain - 2014)

Human Activity Recognition

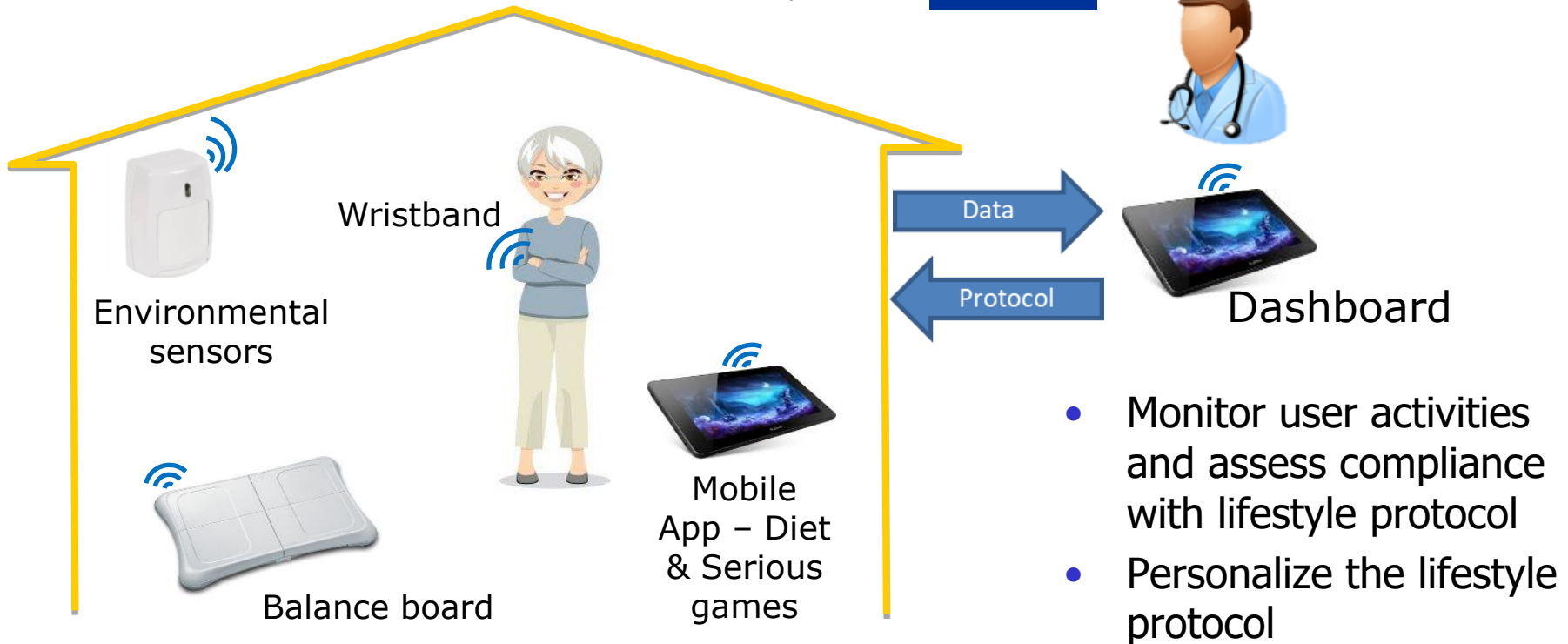


Outputs of ESN Neural Networks (efficient models for temporal data)

Inputs are signals from environmental sensor.

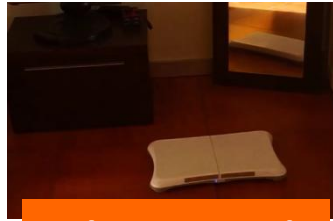
EU FP7

Coordinated by IFC-CNR





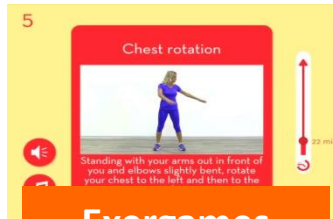
WSN



Balance Board



Bracelet



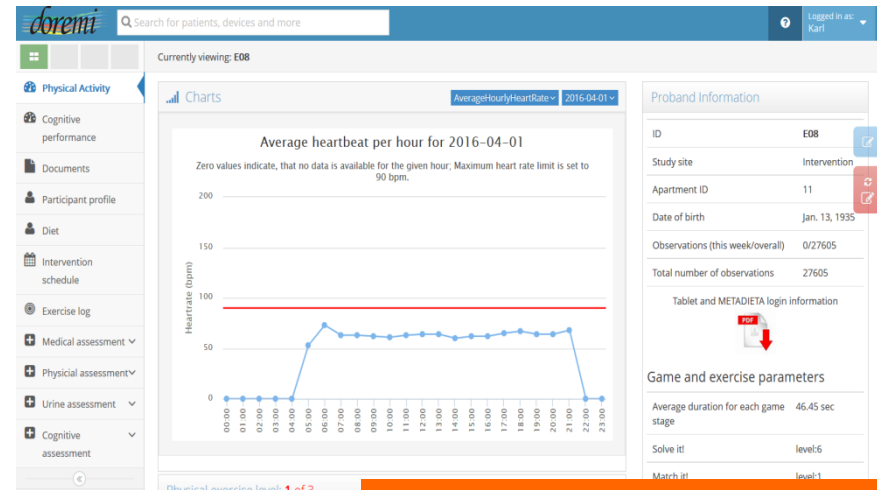
Exergames



Diet App.



Cognitive Games



Web-based Dashboard



feedback





DOREMI (EU FP7)



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- Analysis of dynamical/temporal sensor data for predictive modeling of environmental context and user activities
- **Recognition of behavioral pattern for medical purposes, e.g.:**
 - Balance assessment
 - Level of physical activities/ Kcal energy consumption
 - Social interactions

from accelerometers and heart rate by a user bracelet and environmental sensors (real-world data!)



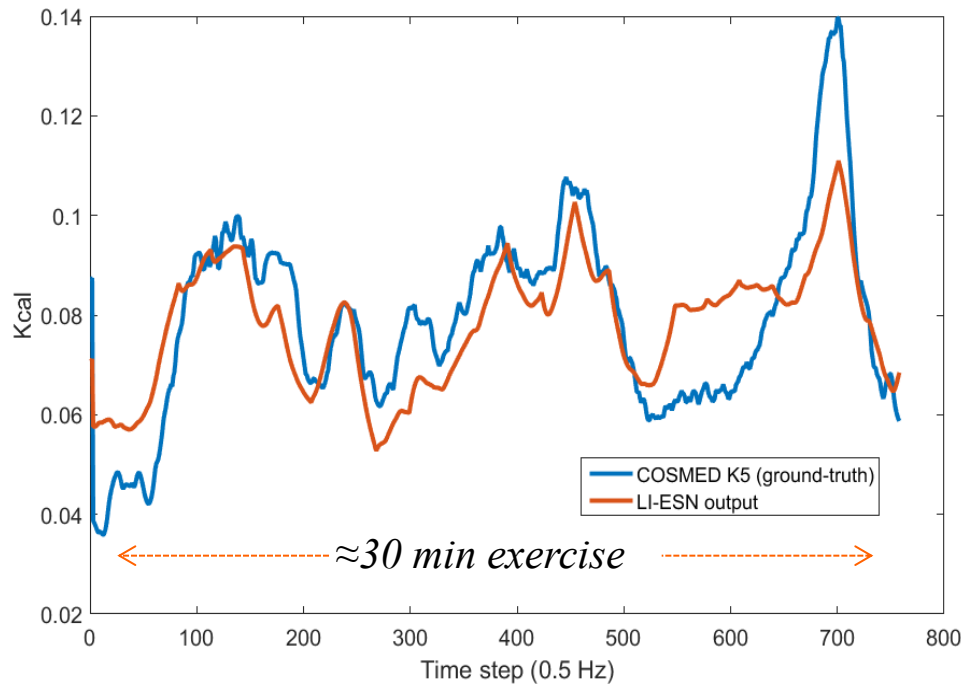
DOREMI (start nov. 13): Decrease of cognitive decline, malnutrition and sedentariness by elderly empowerment in lifestyle Management and social Inclusion



Physical Activity Expenditure Estimation

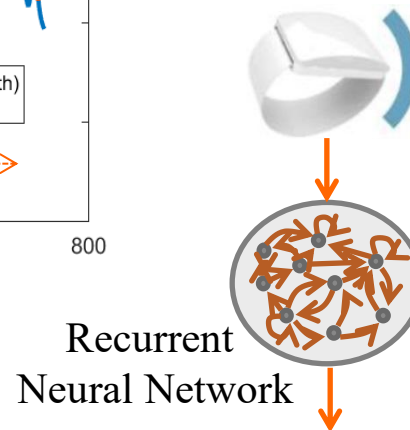
Automatic system for estimating calories consumption from heart rate (HR) measurements (by a bracelet)

■ Ground-truth calories consumption (Kcal) from COSMED K5



Target vs Output on a test sequence

- Improves literature approaches
- Tuned for elderly (by learning on a group of volunteers)
- Dramatically improves the accuracy of commercial devices (≈ 20 times better)



VS



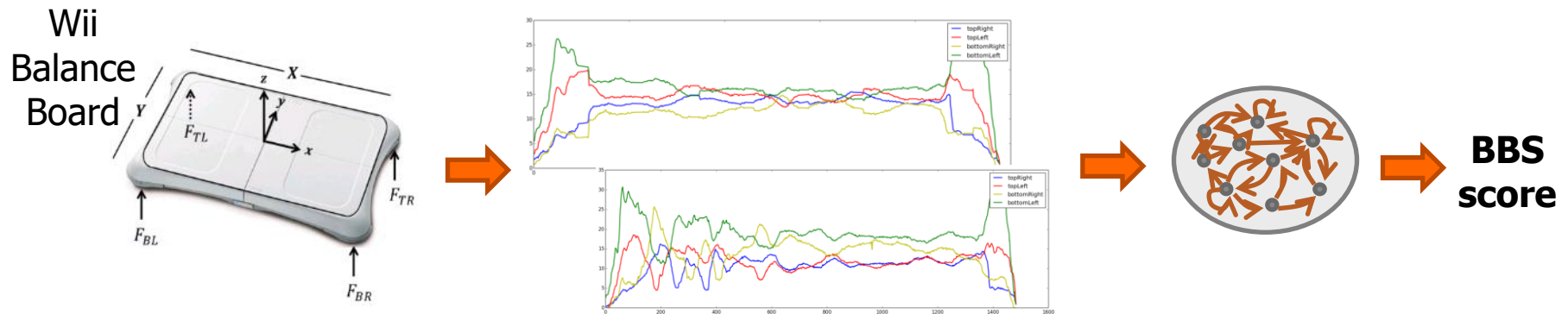
Balance Skills Estimation



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An **unobtrusive automatic system** for **balance assessment** in elderly

- Berg Balance Scale (**BBS**) test: 14 exercises/items (~30 min.)
- Autonomous, unobtrusive & simple: the patient can use it even without supervision by clinicians
- **Save time** for monitoring balance stability (1 vs 14 ex.): 10 sec.
- **Recurrent Neural Networks** for time-series allow to automatically **exploit the richness** of the signal dynamics



Protection of animal populations (species preservation)



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- Pattern recognition on tortoise activity (stream from sensor data) in natural parks (Unipi)



Localizing tortoise nests by neural networks
PLOS ONE, 11(3), 2016.

Tortoise@ recognizes the digging phase using neural networks

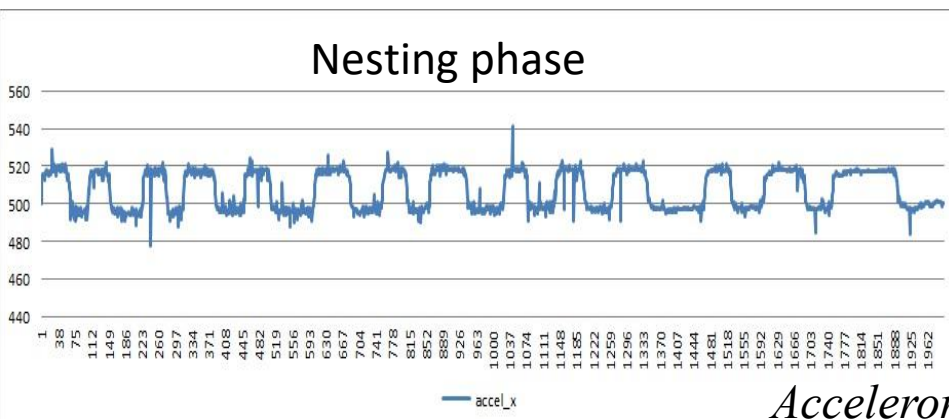


Eating phase

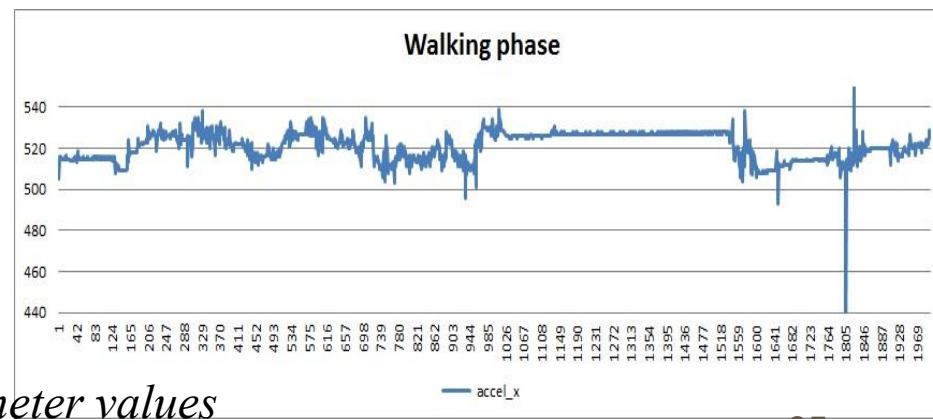


Walking phase

Nesting phase



Walking phase



Accelerometer values

The final result ;-)

A device with a ML classifier identifies the excavation activity of the tortoises.



When the excavation phase is recognized, the device sends a message to the herpetologists.



The herpetologists take the eggs and protect them in a protection center.



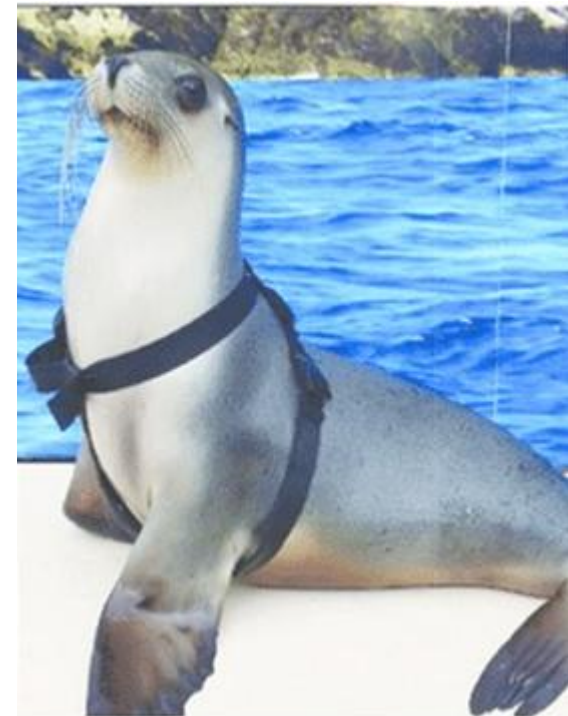
But also....



Penguins

With Australian universities collaborations

Seals

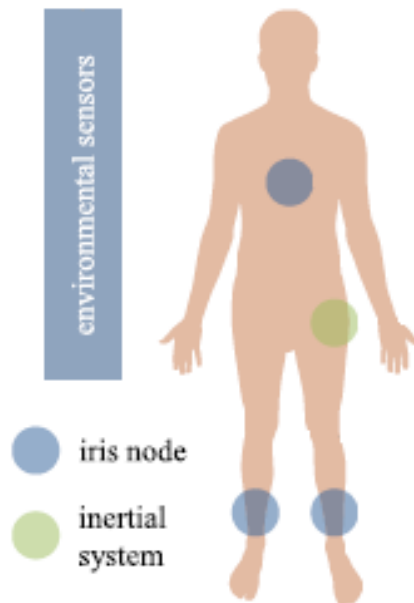


And for Humans ...



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- Human Activity Recognition by ESN



- **Classification of human daily activities** from RSS data generated by sensors worn by the user
 - **Input:** temporal sequences of RSS values (6 dimensional vector for each time step, noisy data)
 - **Target:** classification of human activity (bending, cycling , lying, sitting, standing, walking)
 - Extremely good accuracy ($\approx 0,99$) and F1 score ($\approx 0,96$)
 - 2nd Prize at **2013** EvAAL International Competition



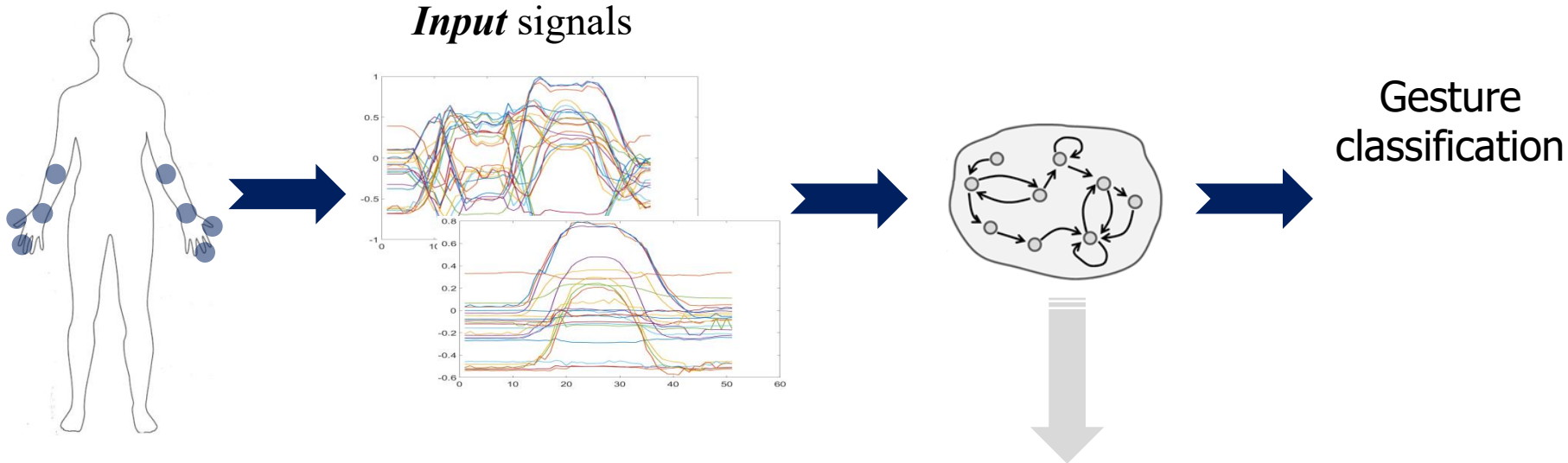
Dataset is available online on the UCI repository

<http://archive.ics.uci.edu/ml/datasets/Activity+Recognition+system+based+on+Multisensor+data+fusion+%28AReM%29>

Human Gesture Recognition by ESN

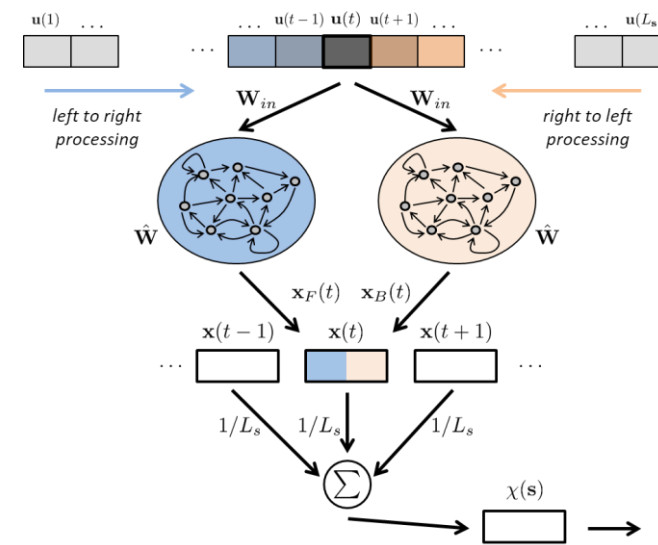


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- Experimentally assessed in the AALTD2016 International challenge (ECML/PKDD**2016**)

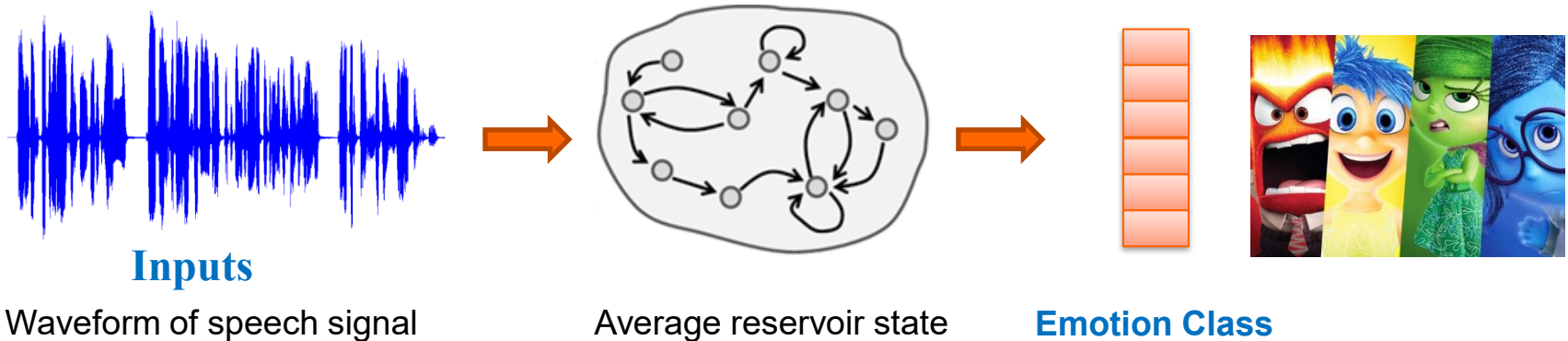
Validation Set	Test Set
97.8%	94.4%



Emotion Recognition from Speech by ESN



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- Speech Processing
- **EVALITA 2014 – Emotion recognition track (Sentiment Analysis)**
- **INPUT:** Time-series representing the waveform of speech directly (no explicit feature extraction required)
- **OUTPUT:** emotion (anger, disgust, fear, joy, sadness, surprise, and neutral)
- Efficiency and simplicity of the system
- Promising performances achieved

Neonatology

A **machine learning** approach to *estimate preterm infants survival*

- **Aim:** *Early estimation of mortality risk* of very preterm neonates
 - from features such as birth-weight, gestational age, sex and other physiological variables (12) measured in the first minutes
- Allow to help doctor decision for immediate therapy, to better advise parents, etc.
- Data set of more than 28000 infant records (from the Italian Neonatal Network)
- Relevance of an “**open data**” approach!
- **On-line** PISA (Preterm Infants Survival Assessment) predictor!
[<http://pisascore.itc.unipi.it/>]



A machine learning approach to estimating preterm infants survival: development of the Preterm Infants Survival Assessment (PISA) predictor

Scientific Reports - Nature (2018) vol. 8:13743.

Unipi main page 16 Oct 2018



UNIVERSITÀ DI PISA

IT EN 中文 ES PT RU

ATENEOSTUDENTIRICERCAINTERNAZIONALETERRITORIO/IMPRESERPERSONALE



Il Monte Pisano tra i "Luoghi del Cuore"

L'Ateneo sostiene la candidatura per la campagna del FAI

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STUDENTI

- » Matricolandosi
- » Corsi di Laurea
- » Master
- » Dottorati
- » Scuole di specializzazione
- » Summer Schools
- » Formazione insegnanti
- » Segreterie
- » Alice -Servizi online
- » Orientamento
- » Borse di studio e alloggio
- » Tirocini e Career Service
- » USID - servizi per disabili

RICERCA

- » Bandi ricerca
- » ARPI Archivio della Ricerca
- » Trasferimento tecnologico
- » Strutture e facilities

PERSONALE

- » Docenti
- » Tecnici e amministrativi
- » Stipendi e pensioni

UNIPIEVENTI

- 16 Ottobre**

Thousands are Sailing
Mostra di foto d'epoca per il progetto Europea Migration
- 16 Ottobre**

Ludoteca Scientifica

UNIPINEWS



Dall'Intelligenza Artificiale uno strumento per valutare la sopravvivenza dei neonati prematuri

Si chiama PISA ed è stato realizzato dagli informatici dell'Ateneo e i neonatologi dell'Ospedale Versilia

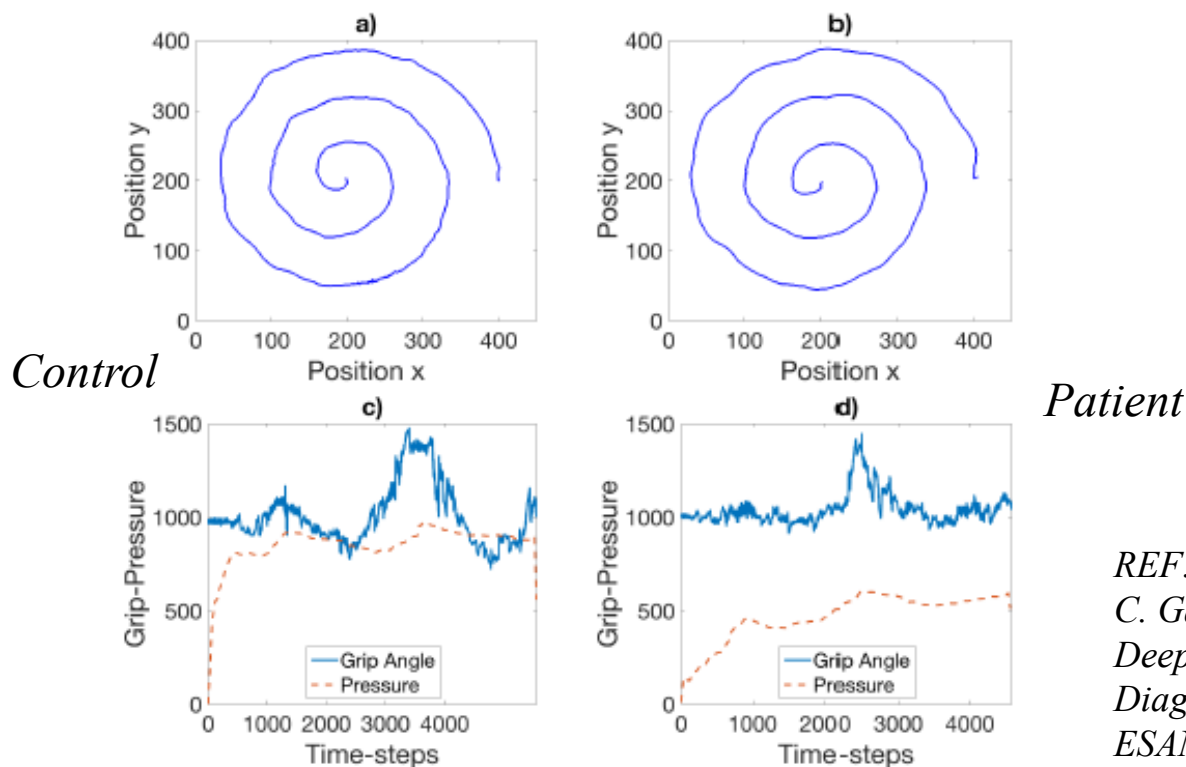
[Vedi tutto \[English version\]](#)





Parkinson

- The identification of Parkinson diseases is performed by analyzing the **time-series** collected from a tablet device during the sketching of spiral tests
- by a **deep recurrent neural networks** (Deep ESN)

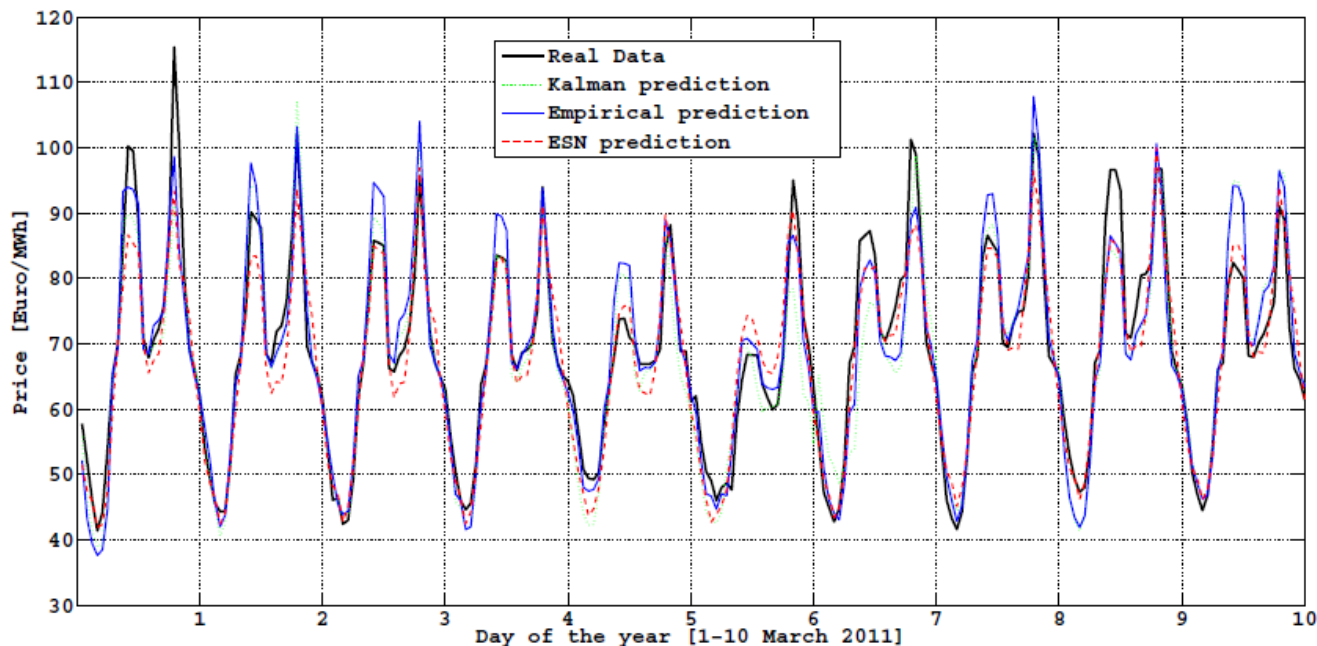


REF:

C. Gallicchio, A. Micheli, L. Pedrelli.
*Deep Echo State Networks for
Diagnosis of Parkinson's Disease.*
ESANN April 2018

Applications of ESN for smart grid applications

- Prediction of the Electricity price on the Italian Market
 - one day-ahead hourly electricity price forecast: most accurate price predictions imply significant economic savings in many smart grid activities



Accurate prediction of hourly electricity price (less than 10% MAPE error)

Earthquake PRJ



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- **ASPIS** PRJ with INGV: Istituto Nazionale di Geofisica e Vulcanologia also CS and Physic dep. + 3 companies
- «*Allerta sismica precoce per infrastrutture sensibili*»
 - EN: Early seismic alert for sensitive infrastructures
- VIRGO @ EGO (European Gravitational Observatory).



The **Virgo** (3-km long) interferometer antenna is designed to *detect gravitational waves* predicted by the general theory of relativity.
Cascina, Pisa.

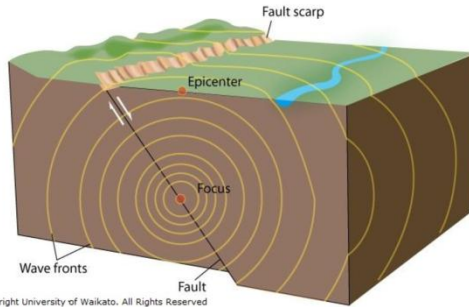
- But early detection can be useful in many other scenario (to save human life!)

Example: seismic waves



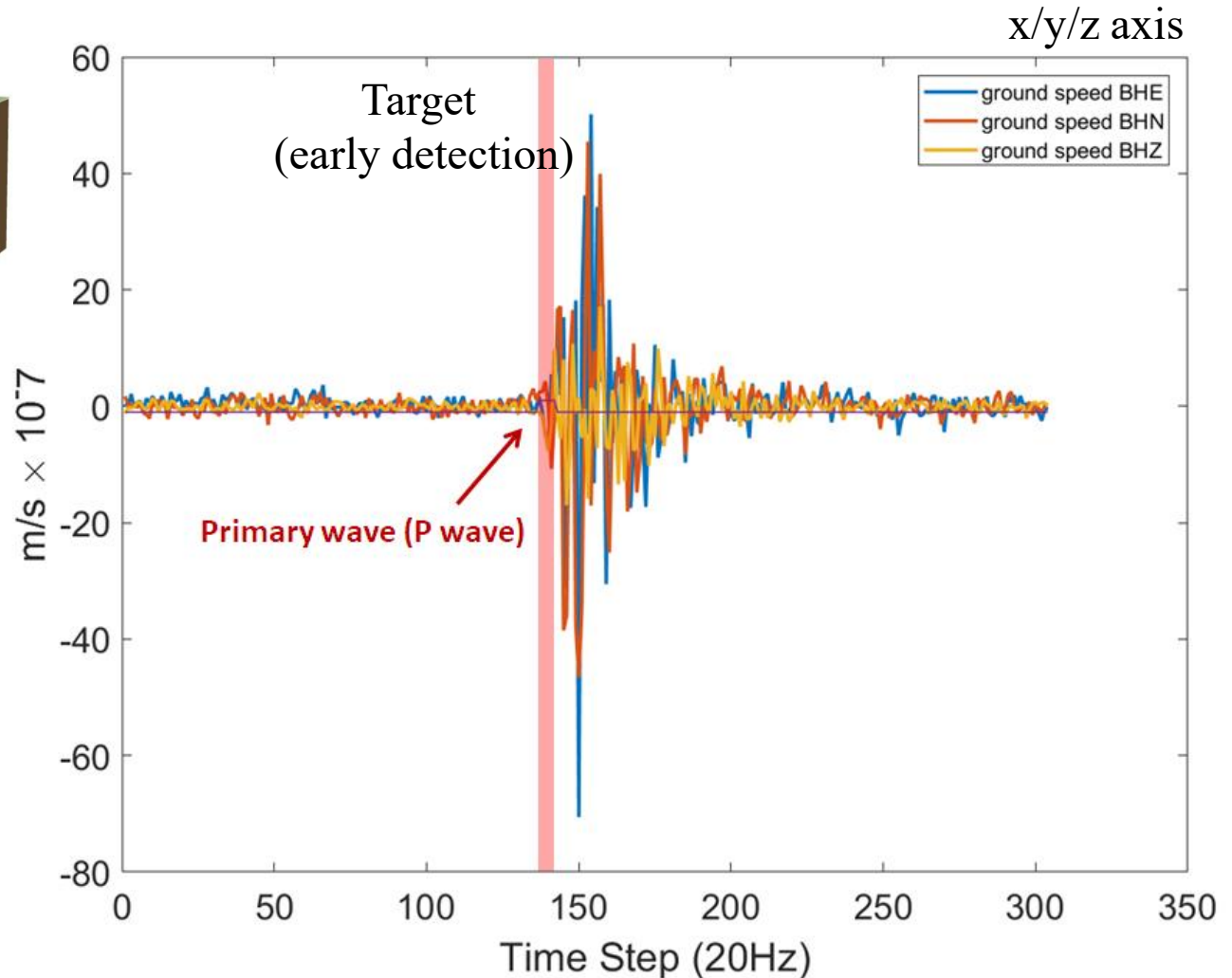
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Seismic Waves Radiate from the Focus of an Earthquake



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**Thesis are
still available**



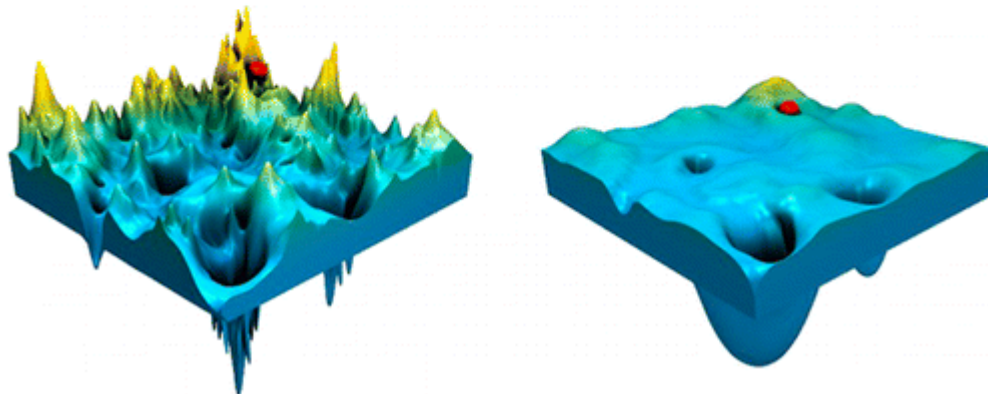
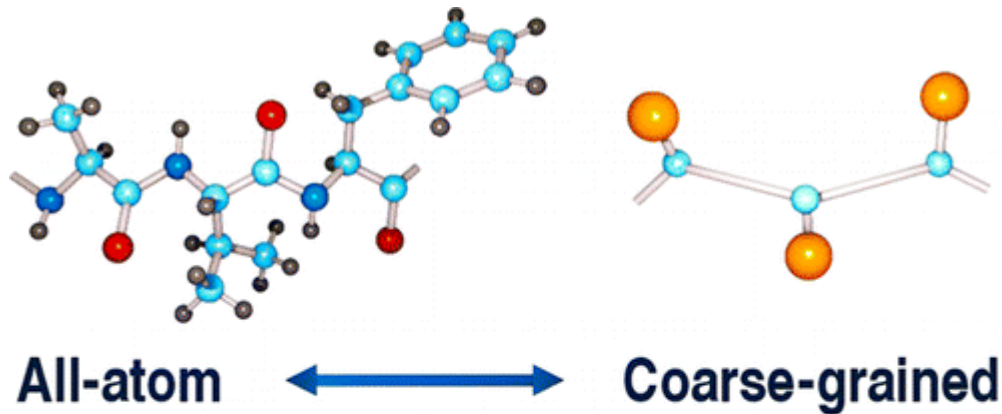
Trees: ITA Emoji 2018

- Emoji Prediction via Tree Echo State Networks
- Italian tweets → Parser → Tree (grammar relations and word embedding) → TreeESN → Emojis class

				
20.27%	19.86%	9.45%	5.35%	5.13%
				
4.11%	3.54%	3.33%	2.80%	2.57%
				
2.18%	2.16%	2.03%	1.94%	1.78%
				
1.67%	1.55%	1.52%	1.49%	1.39%
				
1.37%	1.28%	1.12%	1.07%	1.06%

Emojis under consideration and their frequency within the dataset.

Graphs: Coarse grained models for Proteins (Biophysics with Univ. of Trento)



By NN for graphs!



NEW PROJECTS (on going): and Thesis proposals



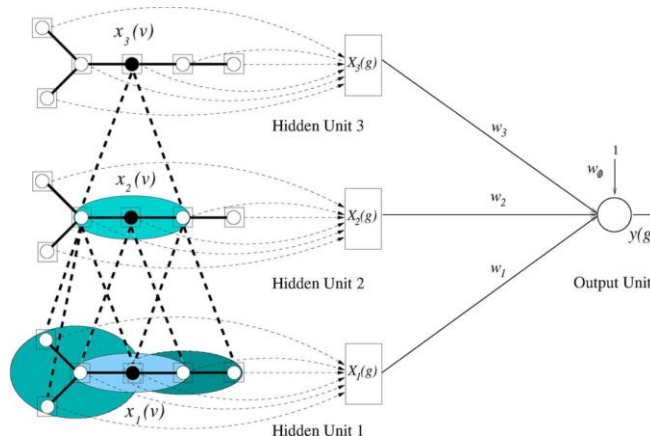
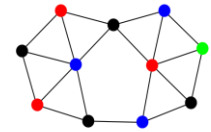
Main on-going projects and developments



*“tie a knot in your
handkerchief”*

Model developments (I)

- **ML/Deep NN for graphs** area: see the SDL lecture and its bibliography, e.g.:
 - Constructive approaches by *Deep NN*: **NN4G basis**
 - E.g. heterogenous pool in the cascade correlation construction



A. Micheli. **Neural network for graphs**: a contextual constructive approach, *IEEE Transactions on Neural Networks*, 20 (3): 498-511, **2009**

➤ **Thesis are ready to start**

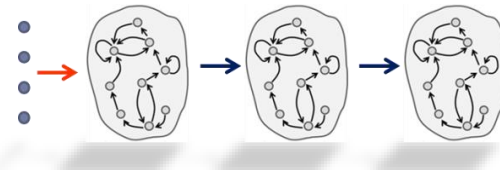


Many new variations are possible following the current developments and our proposals aiming at involving **you** in papers on top-tier venues for ML (ICML, AAAI, NeurIPS and Journals)

Model developments

- **RC/ESN area**

- Analysis of basic properties and architectures (typically simplified architectures, like rings)
- **Deep ESN**: further advancements (typically simplifying the architecture)
- **Extensions to SD (trees, graphs)**: further advancements
 - (e.g. properties analyses, comparison with trained GNN models, sw ...)
- **New: over-smoothing issues for graphs** → See lect. On SDL learning with references, e.g. Micheli, Tortorella. *Addressing hererophily with graph echo state networks*. Neurocomputing, 2023
- **New: temporal Graphs** by Dynamic-GraphESN & **Continual Learning** → See lect. On SDL learning with references, e.g. Micheli, Tortorella. *Discrete-time Dynamic Graph Echo State Networks*. Neurocomputing (2022)



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Many new variations are possible following the current developments and our proposals aiming at involving **you** in papers on top-tier venues for ML (ICML, AAAI, NeurIPS and Journals)

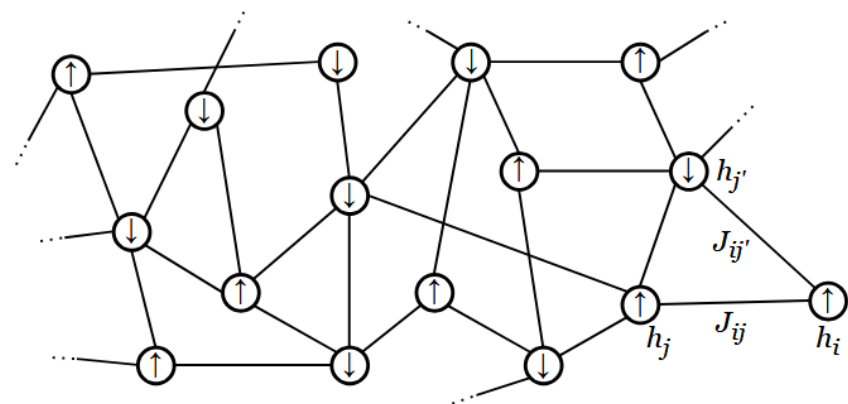


Another recent example: Ising Models for Graphs



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- Objectives
 - Efficient HW implementation of Graph RC
- Means:
 - Node representations as Ising spin configurations
 - Sampled from Boltzmann distribution (energy minima)
- How can we implement it?
 - Non-conventional hardware
 - Spintronics, optoelectronics, ...
- Who: Collaboration with... TDB



$$\mathcal{H}(\mathbf{h}) = -\mathbf{h}^T \underbrace{(\mathbf{A} \otimes \widehat{\mathbf{W}})}_{\mathbf{J}} \mathbf{h} - \mu \underbrace{(\mathbf{W}_{\text{in}} \mathbf{x})^T}_{\mathbf{b}^T} \mathbf{h}$$

Spin coupling (according to the graph topology, see the fig.)

External magnetic field

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Tortorella, Brau, Micheli, “[Encoding Graph Topology with Randomized Ising Models](#)”
ESANN 2025 Proceedings, pp. 271–276.

An applicative example: Drug Discovery via Graph Learning & Graph XAI

- Motivations:

Discovering new drugs requires:

- Finding new molecules with specific target activity
- Lab trials are too costly

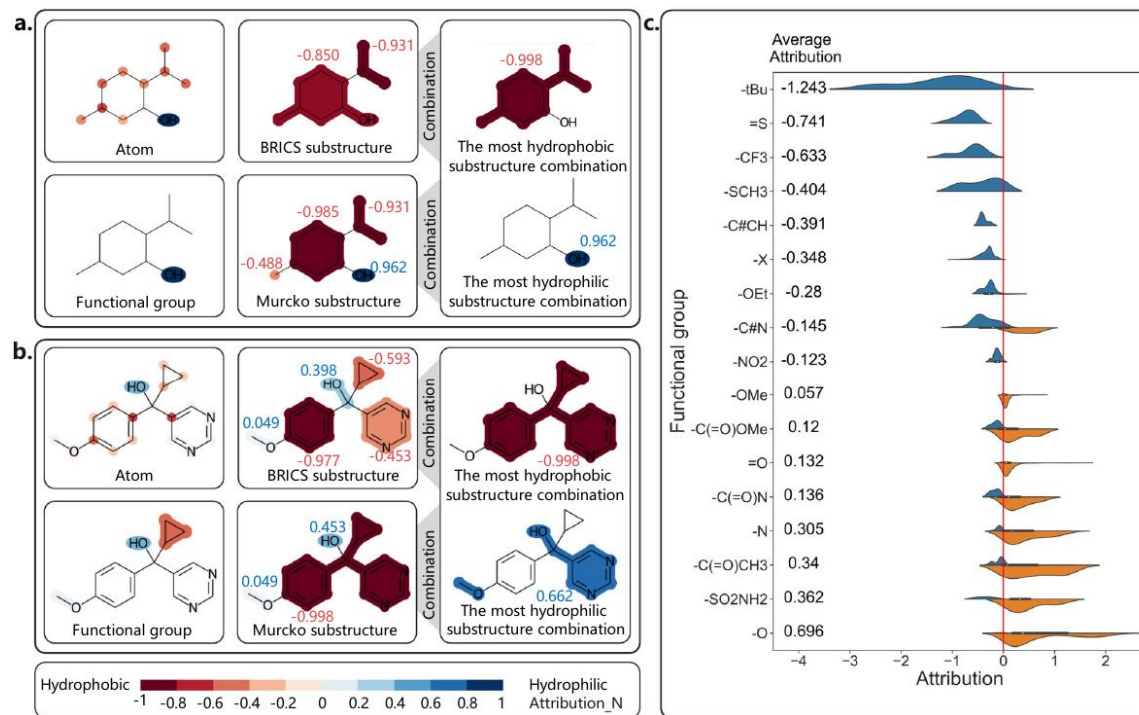
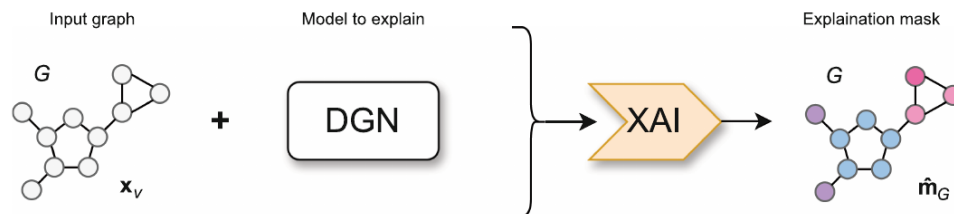
- Objectives:

- Learning for graphs to support Drug Discovery process
- Why should a molecule be a potential drug?

- Who

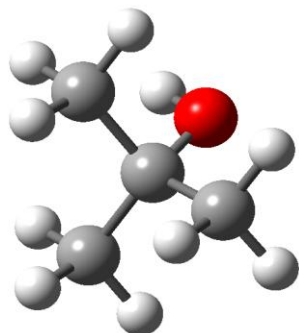
- Collaborations with Helmholtz Munich (Institute of Research)

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Explaining learning model predictions

Cheminformatics & Toxicity



QSPR: Correlate chemical structure of molecules with their properties

**Property Value
e.g Toxicity:**

$\left\{ \begin{array}{ll} +1 & \text{skull and crossbones} \\ -1 & \text{smiley face} \end{array} \right.$

- Means: again,
 - Learning for graphs
 - Graph XAI
- Who
 - Collaborations with *Mario Negri Institute* of Pharmacological Research - Department of Environmental Health Science

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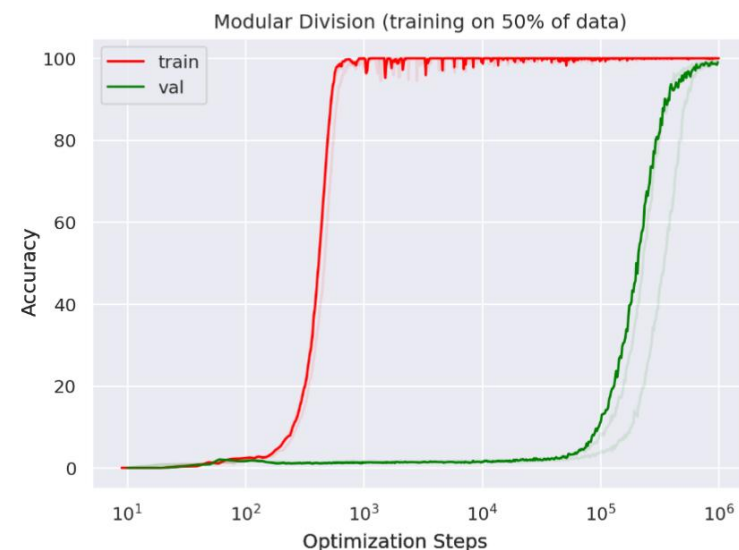
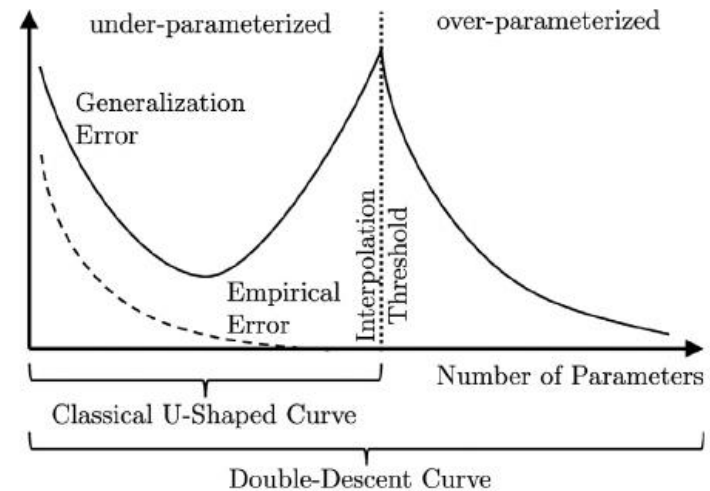
***Thesis* related to NN for graphs and Graph XAI**



Theory: Investigating Grokking and Double descent

- Motivations
 - Some learning models do not follow the classic underfitting/overfitting generalization curve
- Objectives
 - How can we explain it?
- Means
 - Model capacity measures beyond VC-Dim
 - Advancing learning theory
- Who
 - Collaborations with UniGe

➤ **Thesis are ready to start**



PNRR Project NextGenerationEU



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- **FAIR** – Future Artificial Intelligence Research is the Italian project approved in the call “Extended Partnerships: **Artificial Intelligence. Fundamental aspects**” of the NRRP.
- 3 years, started on January 2023
- The FAIR project brings together 350 researchers and is developed in 10 spokes
 - Spoke 1: *Human-centered AI* (Leader: **UNIFI**), e.g.:
 - Objective 1: “human-in-the-loop” machine learning and reasoning
 - WP1.3 - Human centered *Lifelong Learning for Complex Data*



FUTURE AI RESEARCH



WP1.3 FAIR Objectives

WP1.3 - Human centered Lifelong Learning for Complex Data

Lead: UNIFI, PIs: **Alessio Micheli, Fabrizio Sebastiani.**

- study how deep learning architectures (transformers, graph NN, GANs, autoencoders, ...) can **continuously and incrementally learn new concepts**, considering also the “human in the loop” and small/scarce data.
- push forward the frontier of **learning in complex dynamic domains** (streams; time series; trajectories; graphs; knowledge graphs; multimedia, time-varying networks)
- **Cross-modal/cross-language lifelong learning**
- Other Key Personnel: Davide Bacciu (Unipi), Vincenzo Lomonaco (Unipi), Alessandro Lenci (Unipi), Fabrizio Falchi (CNR), Gualtiero Fantoni (Unipi)

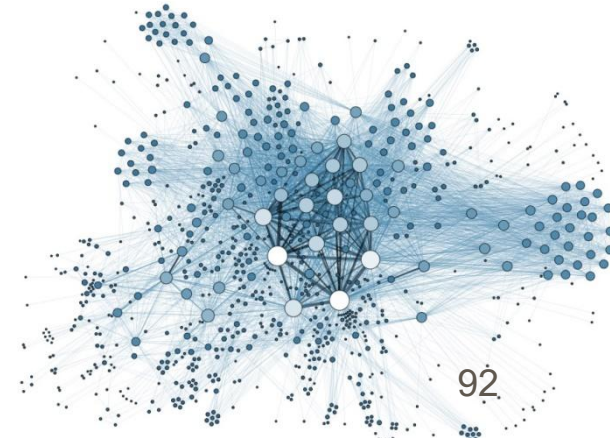
Trustworthy for graphs



Trustworthy AI for complex data (structured domains):

New models development and analysis considering

- **Explainability**
 - Interpretate the outcome of a NN for graphs, e.g. in the molecular domain, why this drug his effective?
 - Or connecting to knowledge graphs, bridging symbolic and sub-symbolic reasoning, etc.
- **Robustness**
 - Perturbations and adversarial attacks to graph NN
 - Difficult: effect small perturbations on graph structure (e.g. to induce to misclassify a person in a social network). It can be useful also for medical data analysis, including COVID data.
- **Fairness and privacy preserving**
 - e.g. in social data analysis and again COVID data
- **NEW:** Integration with the **continual learning**

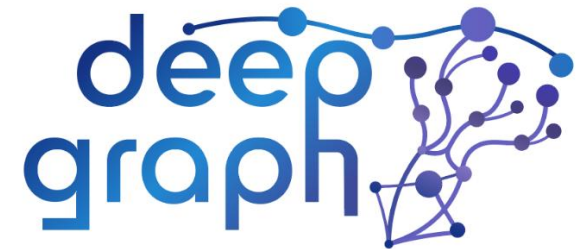


On going research: Deep Graph

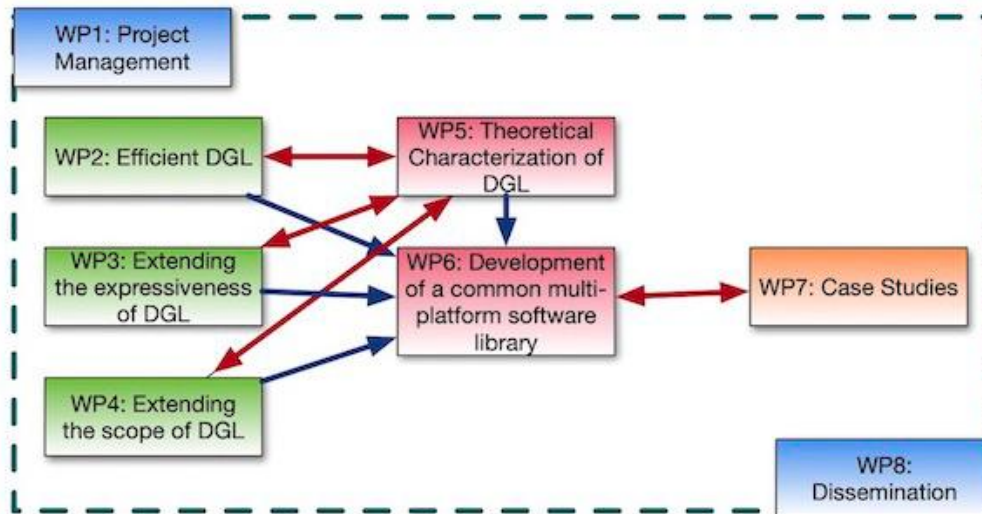


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- PRIN (national project), 2Y, **starting Feb. 2025**
- DEEP-GRAPH: Design and Theory of Deep Graph Learning
 - University of Padova (Prof. A. Sperduti)
 - University of Pisa (Prof. A. Micheli)
 - University of Siena (Prof. F. Scarselli)



Ministero
dell'Università
e della Ricerca



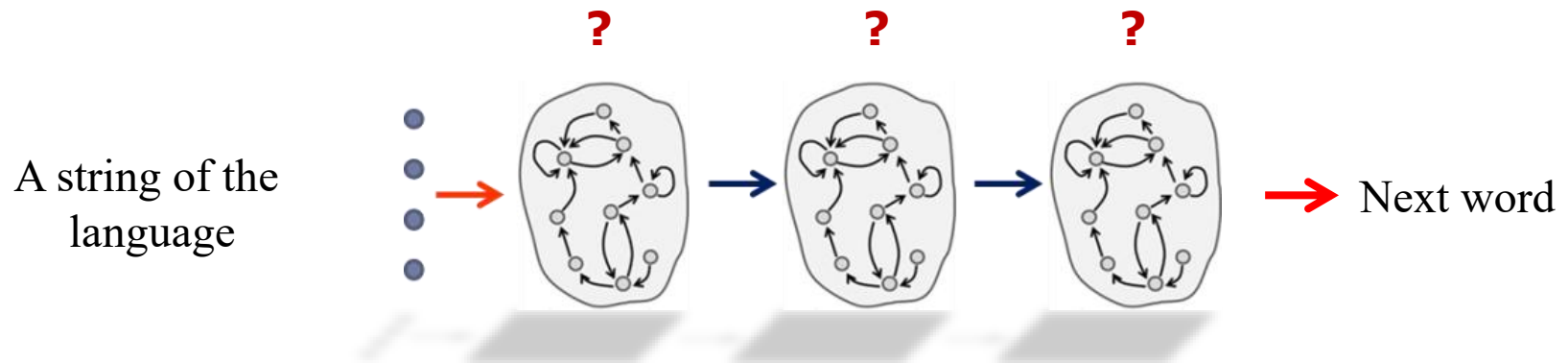
**Thesis are
ready to start**



Further Developments and Applications

NLP related

- Trade-off accuracy/efficiency using RC approaches for NLP (vs LLM)
- Analysis of the representations developed in the internal layers of deep ESN (RNN)
 - I.e. analyses of different levels of abstraction for language encoding
- With Computational Linguistic researchers



**Thesis are
ready to start**



A recent project in Cardiology: BrAID



BrAID Project

Brugada syndrome and Artificial Intelligence applications to Diagnosis

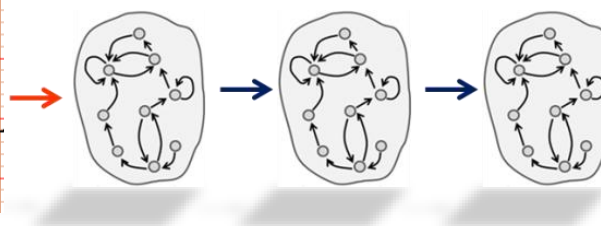
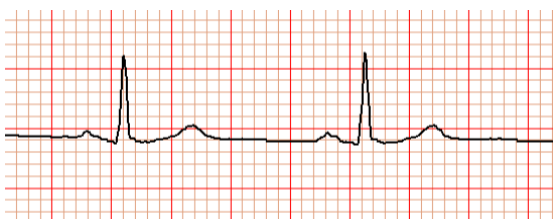


Regione Toscana

braid

- Brugada syndrome (BrS): not common but dangerous disease (sudden cardiac death)
 - No symptoms
 - BrS is diagnosed by identifying **characteristic patterns** on the electrocardiogram (ECG)

ECG (electrocardiographic) signal



Brugada **diagnosis alert**:
allowing advantages in term of
screening automatization



Results: accuracy over 90%, w.r.t. 88.0% by clinicians (blind test)

Vozzi, Pedrelli, Dimitri, Micheli, et al. Echo state networks for the recognition of type 1 Brugada syndrome from conventional 12-LEAD ECG, Helyon, 2024

BrAID now and next

State-of-the-art:

- We concluded the data collection, so we have now a good dataset with positive (BrS) and negative cases
- We have very good results applying ESN (over 90% of accuracy, with cardilogit at 88)
- We can develop (future projects):
 - **extended data sets** by other collaborations
 - **new approaches** (including the usage of different sources of data, ECG and features, filtered versions, or ECG different leads, omics data, etc.)
 - **new models** (e.g. CNN, RNN, advanced and *hybrid approaches*, interpretable models, comparing with other approaches)
 - compare the results and analysis with **clinical centers** (clinical feedbacks).

**Thesis are
ready to start**



braid

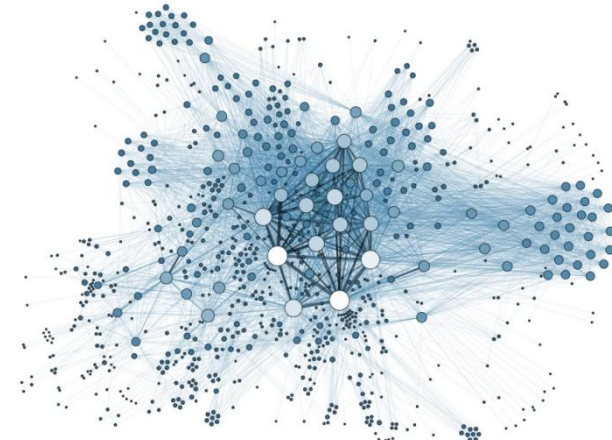
Bioinformatics

Under construction! Ready to apply for **tree/graph data** on:

- System biology (graphs/networks) (with P. Milazzo) *on-going (next slides)*
- Bioinformatics:
 - Bio-molecular Networks → next slides
 - Prediction of proteins function → next slides
 - Pan-genome analysis (by graph representation)



On going: Thesis available!



THE: PNRR Project NextGenerationEU



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- THE- TUSCANY HEALTH ECOSYSTEM

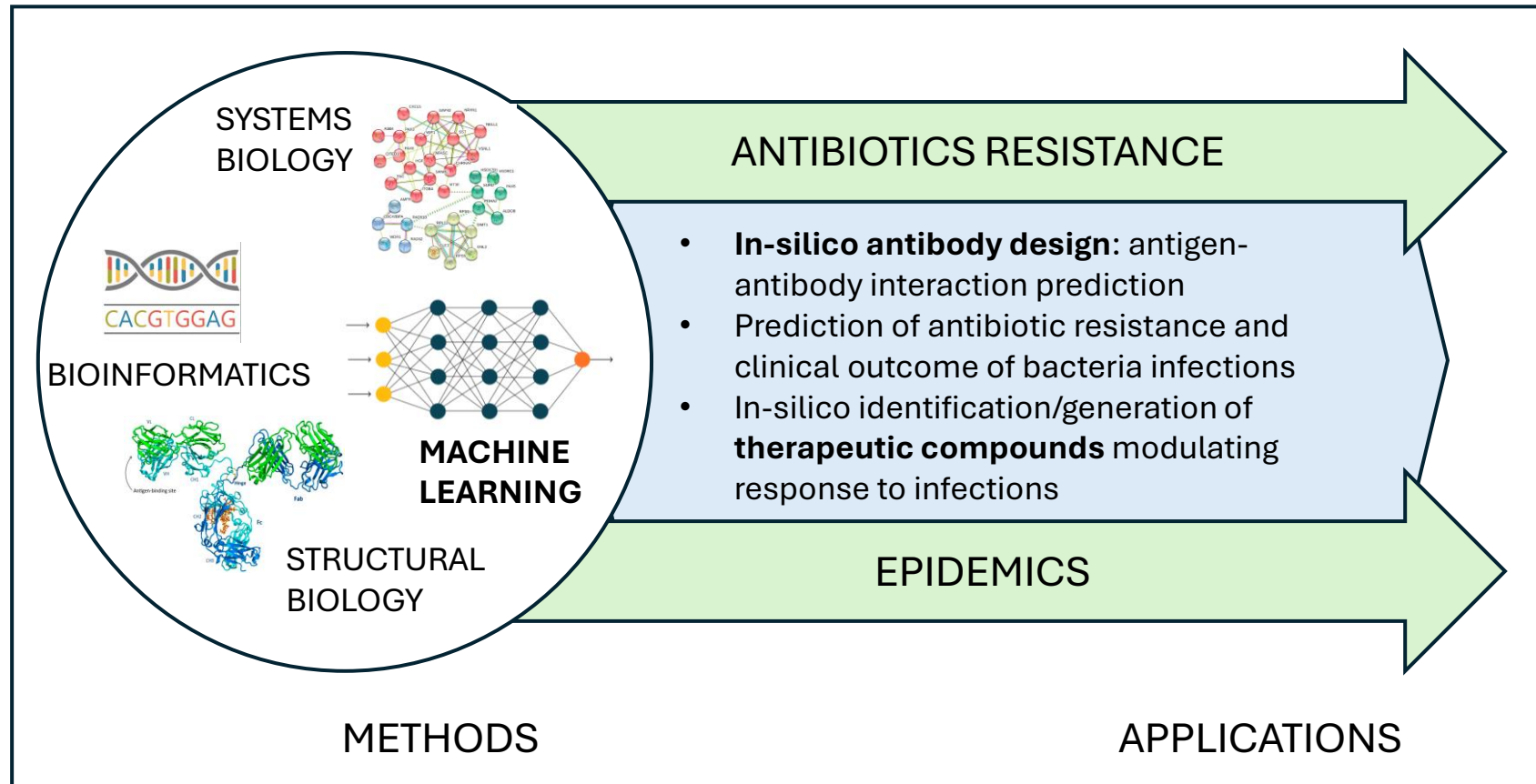
- *Ecosistema dell'innovazione sulle scienze e le tecnologie della vita in Toscana*



- *Spoke: Precision medicine & personalized healthcare*
- System biology, AI and BigData technologies for omics and computational biology
 - Investigation of **machine learning** for structured data in the omics and computational biology domain: *proteins, drugs and genomic structures.*

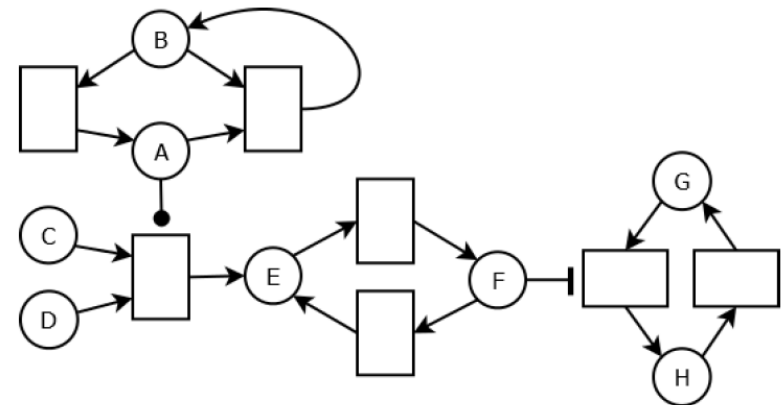


Multidisciplinary and Interregional Research and Clinical Trial Hub for Contrasting Pandemics and Antimicrobial Resistance (PAN-HUB)



Classification of Biochemical Pathway Robustness with Neural Networks for Graphs

- **Biochemical Pathways (BPs)** are sets of chemical reactions among biomolecules
- They model **cell functionality**
- Structure described by **graphs** (e.g. **Petri Nets**), dynamical properties analyzed with Ordinary Differential Equations (ODEs)
- Entities (**nodes**): species (circle) or reactions (square)
Relations (**arcs**): produce, enhance, block
 - there are also Input/Output nodes: induced sub-graphs of paths for different I/O nodes

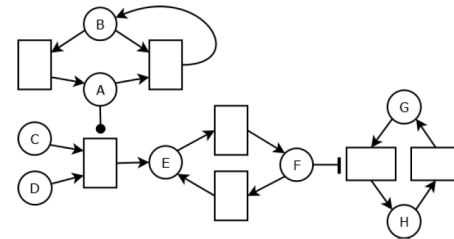


Classification of Biochemical Pathway Robustness with Neural Networks for Graphs



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- **Input:** Pathway Petri nets (their correspondent **directed** graph), for each pair of species in a given net
- **Output:** an associated **robustness** value (0/1)
- Direct prediction for the Petri nets topology, avoiding costly ODE simulations
- Does the networks include enough information to predict robustness? Or other properties?
- Can we extend to Protein-Protein interaction (PPI) networks? (next slide)

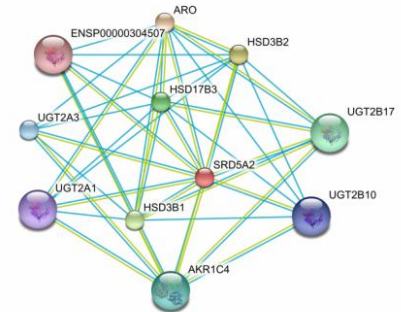


- **Best paper award @ BIOSTEC-BIOINFORMATICS 2020**
Bove, P.; Micheli, A.; Milazzo, P. and Podda, M. (2020). Prediction of Dynamical Properties of Biochemical Pathways with Graph Neural Networks. In Proceedings of the 13th International Joint Conference on Biomedical Engineering Systems and Technologies - Volume 3: BIOINFORMATICS, pages 32-43
- Fontanesi, M., Micheli, A., Milazzo, P., & Podda, M. (**2023**). Exploiting the structure of biochemical pathways to investigate dynamical properties with neural networks for graphs. *Bioinformatics*, 39(11)

Bio-molecular Networks

A Bio-molecular network is a graph representation of

- **Relationships** (of which there are many types) amongst a group of biomolecules.
- **Vertices** or nodes represent biomolecules, including macromolecules such as genes, **proteins**, and RNAs, or small biomolecules like amino acids, sugars, and nucleic acids



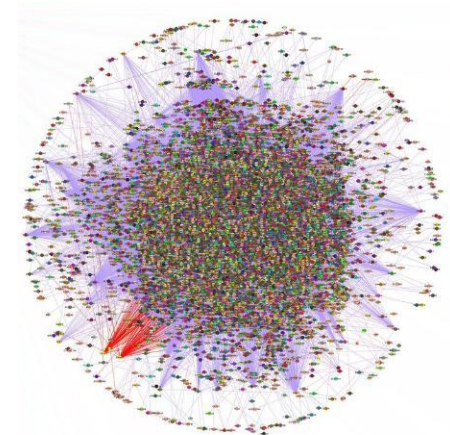
*They can be **very large** (up to millions of nodes)*

In particular, other networks data/problems to extend the application of ML/NN for graphs are:

- **PPI**: Protein Protein Interaction Networks: Proteins (interactions or similarities or ontologies) networks for drug or oncology research
 - These graphs contain a relevant information that can be exploited for conducting different kinds of analysis, such as automatic function prediction, disease gene prioritization, drug repositioning or discovery, etc.

On going: Thesis available!

PPI Networks: first results

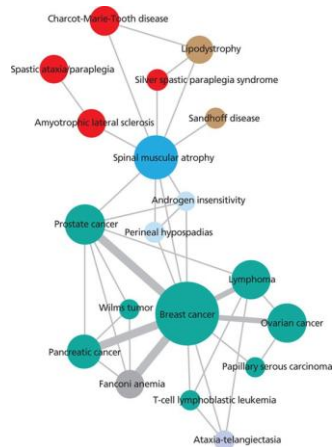


- Protein-Protein Interaction Network (**PPI**)
 - Nodes represent proteins
 - Arcs represent physical associations between two proteins
- **Predicting sensitivity** (perturbation analysis) on PPI
- Starting from biochemical pathways (Biomodels) and mapping to PPI subgraphs (BioGRID, **60K nodes**)
- **Results:**
 - **Efficiency** vs numerical simulation on biochemical pathways (ODE)
 - **High performance** even without protein embeddings, meaning that DGN is relying on PPI topology

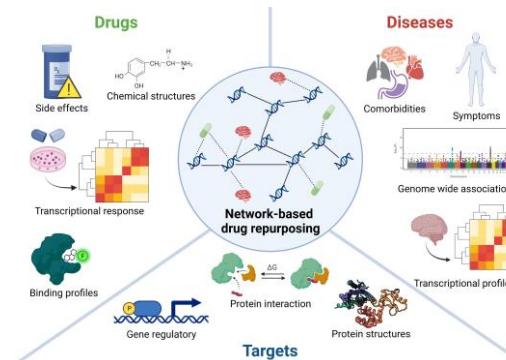
PAN-HUB prj on
antibiotics resistance

Dipalma, Fontanesi, Micheli, Milazzo, Podda. Sensitivity analysis on protein-protein interaction networks through deep graph networks *BMC Bioinformatics* (2025)

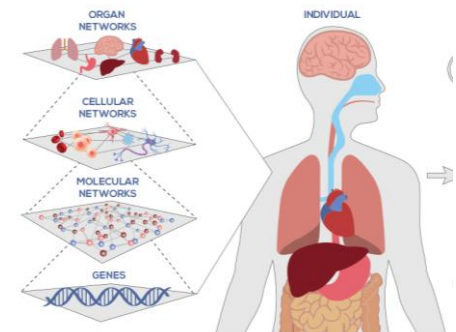
Impacts/Potential usage



Disease understanding



Drug design
and repurposing



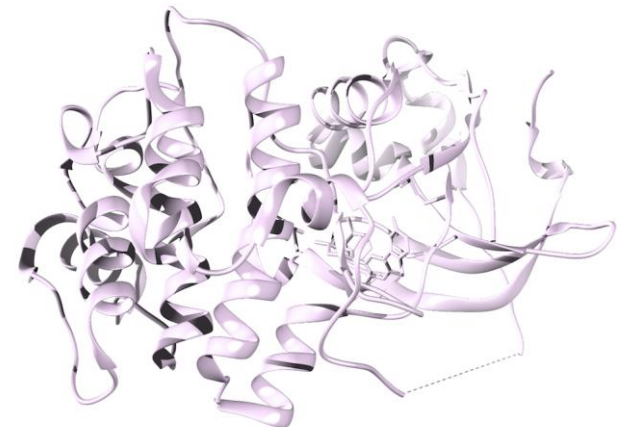
Precision/Personalized
medicine

Prediction of proteins function

- Prediction of proteins function
- Using gene ontology graphs
- Possibility to participate to CAFA challenge
 - There are many proteins in the databases for which the sequence is known, but the function is not.
 - The Critical Assessment of Functional Annotation (CAFA) is an ongoing, global, community-driven effort to evaluate and improve the computational annotation of protein function.

Approaches can include:

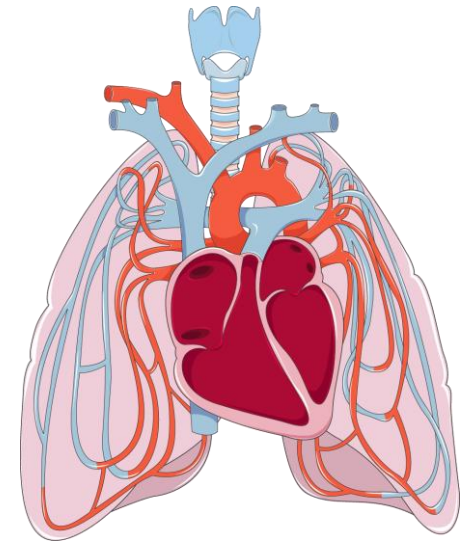
- SVM/**Kernel** building by efficient computation of common sub-graphs between pairs of proteins (with R. Grossi)
- Use of **NN for graphs**



On going: Thesis available!

Pulmonary Arterial Hypertension

- Objectives
 - Predictive risk estimation in pulmonary arterial hypertension
- Means:
 - Classifiers + Survival analysis
 - Importance analysis (XAI) of clinical features (echocardiographic variables)
 - Data sets of clinical data are already available
- Who
 - Collaboration with Prof. Badagliacca, Dep. Clinical Intern, Anesth. and Cardiovascular Sciences, Sapienza University of Rome
 - USA research and clinical centers



On going: Thesis available!

Data Analysis for Industrial Leaders: Example



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- Objectives
 - to develop new approaches for improving the efficiency of urban public transport
- Means:
 - to combine machine learning techniques with combinatorial optimisation methods
- Who
 - Collaboration with Prof. Antonio Frangioni, Andrea Cardia (PhD student), M.A.I.O.R



MAIOR designs and develops software solutions for service planning, resource scheduling and company performance analysis in the transportation industry

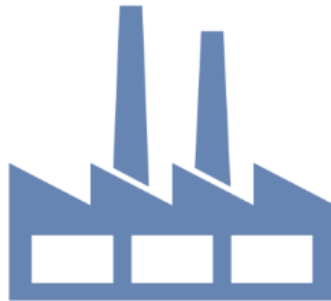
On going: Thesis available!

Data Analysis & ML for Industrial Leaders



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Ongoing requests, stay in touch



On going: Thesis available!

Further projects - Companies

Increasing contacts by companies:

- Keep in touch for developments



Future – just ahead



Material

- Details upon request !!!!
- **Thesis proposal:** *one-to-one discussion*
- Contact me: Alessio Micheli

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For information

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